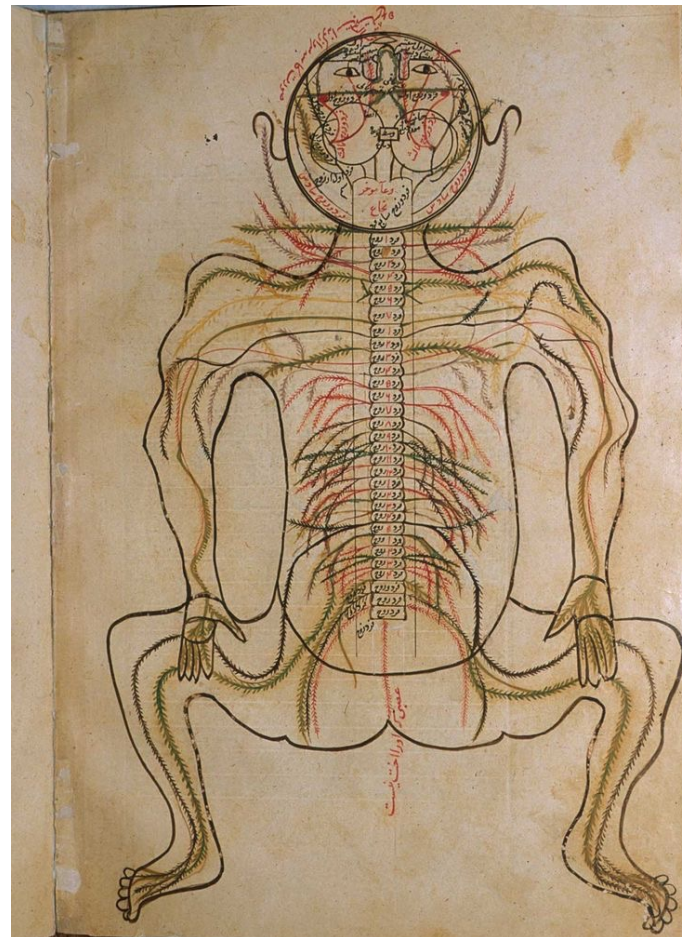


Neurons and nervous system



*Anatomical illustration of the nervous system
by Mansour Ibn-Ilyas, ca. 1400*

Agenda

1. I neuroni
2. Potenziale a riposo e d'azione
3. Accenni allo sviluppo del sistema nervoso:

Struttura del neurone: Golgi e Cajal

Premio Nobel per la Fisiologia e la Medicina (1906)



Camillo Golgi
(1843 – 1926)



Santiago Ramón y Cajal
(1852 – 1934)

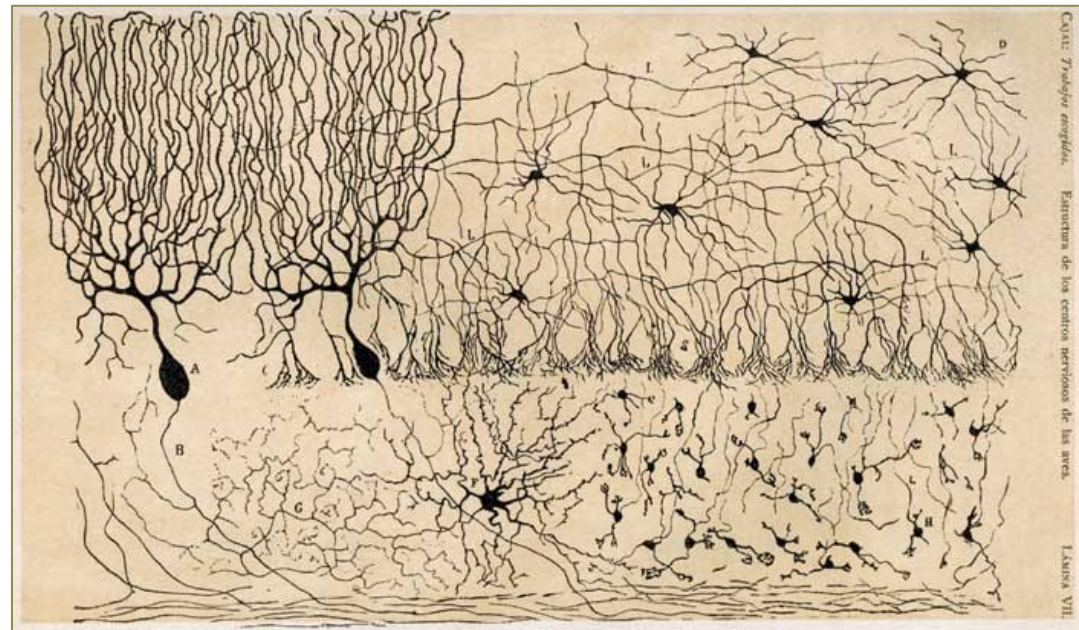
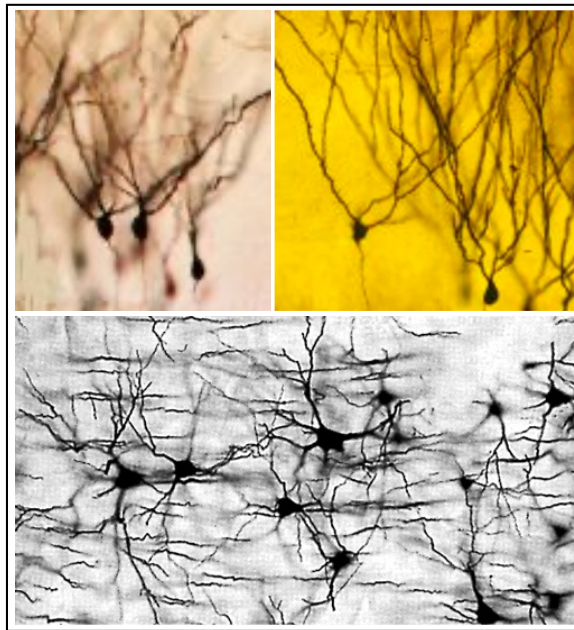
La tecnica di “impregnazione argantica” (colorazione di Golgi - 1873)



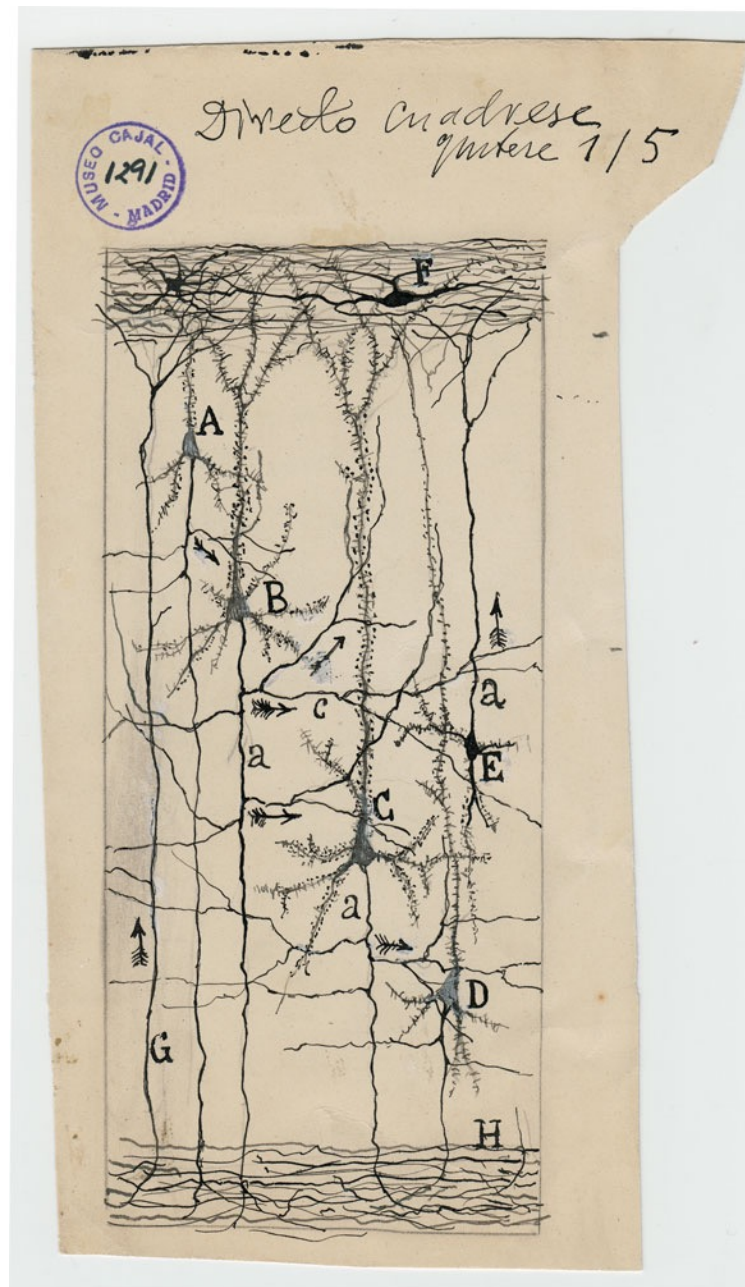
Golgi (1920)



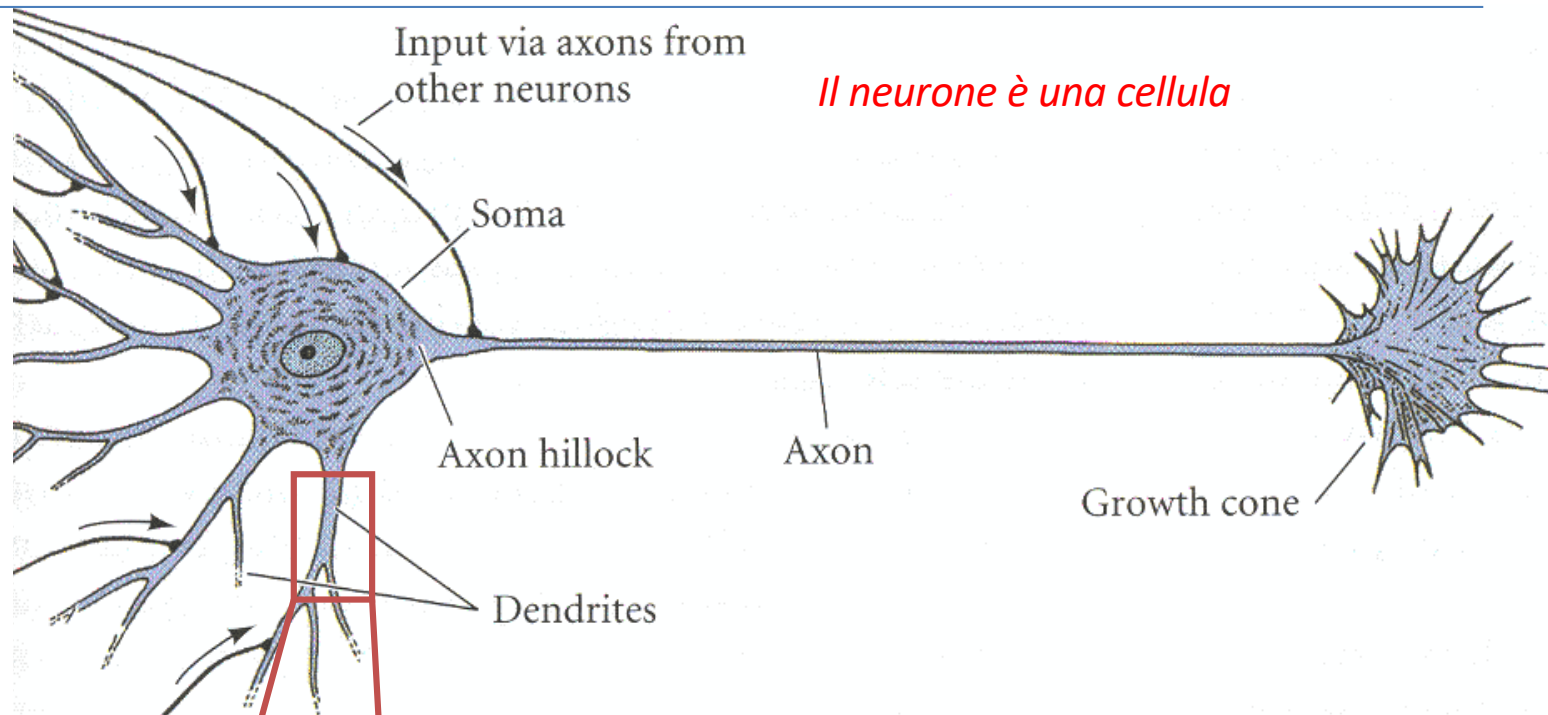
Cajal (circa 1910)



Cajal



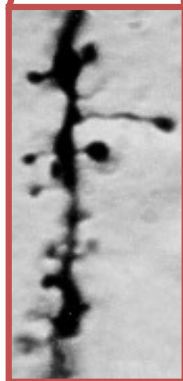
Le “dottrina del neurone” (Waldeyer, 1891 e Cajal, 1906)



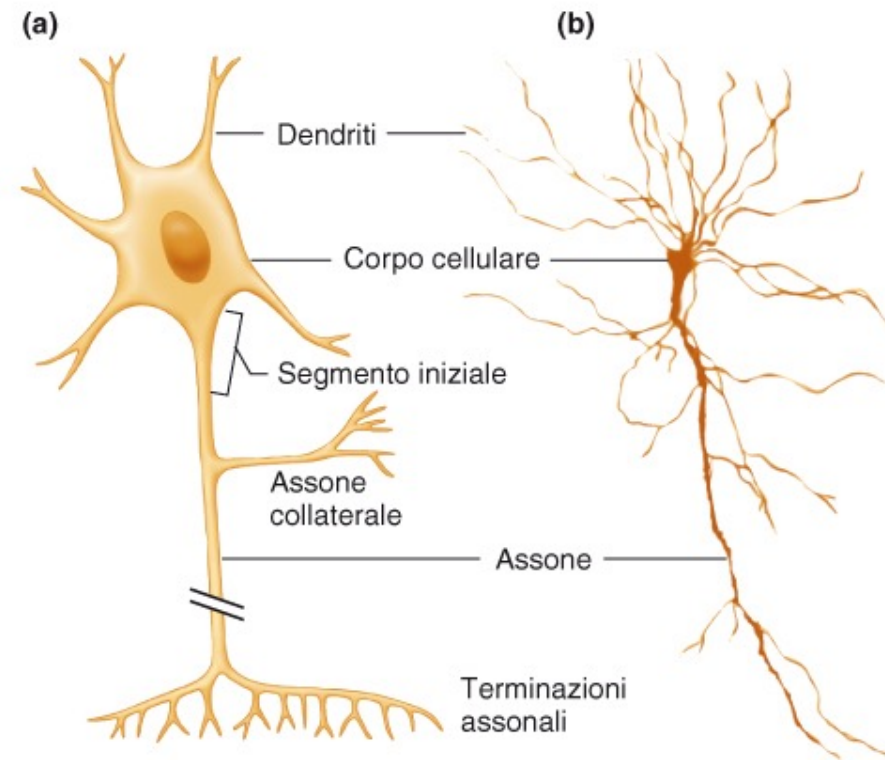
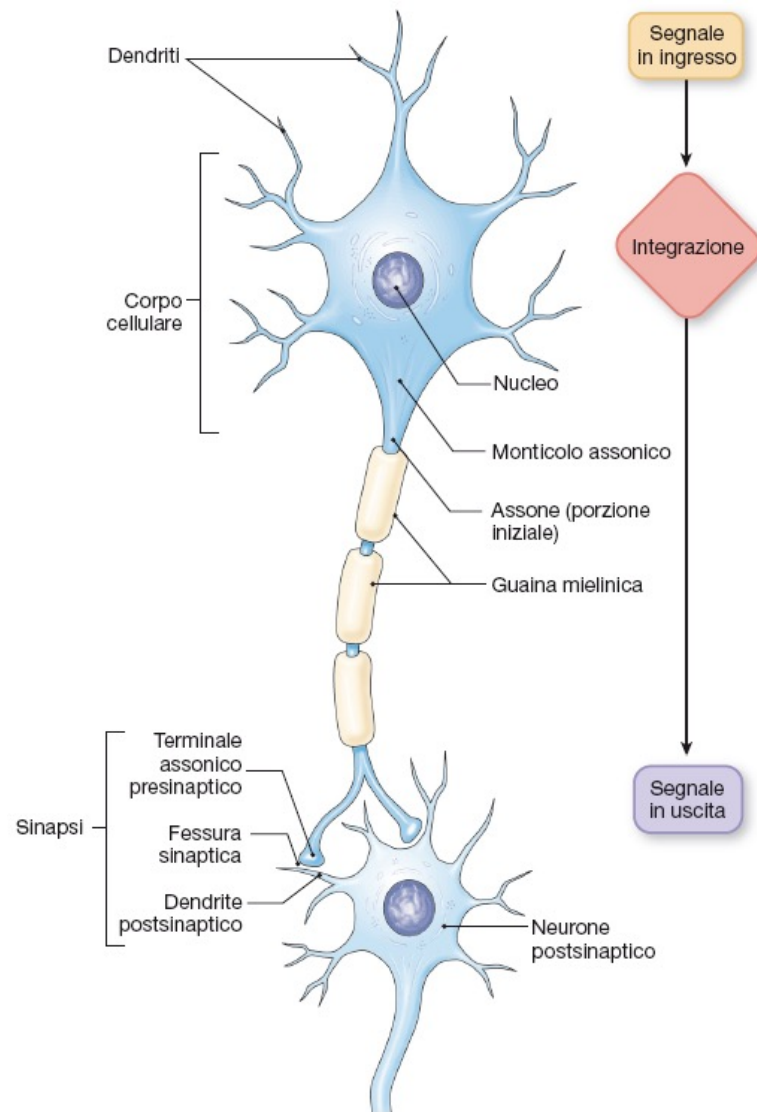
Il neurone è una cellula

Cajal formulò per primo la “legge della polarizzazione dinamica”: i neuroni sono cellule polarizzate, che ricevono informazioni a livello dei dendriti e le trasmettono – anche a lunga distanza – attraverso l’assone.

Cajal descrisse per primo (nel 1891) anche le “spine dendritiche”, ovvero i punti in cui si formano le sinapsi

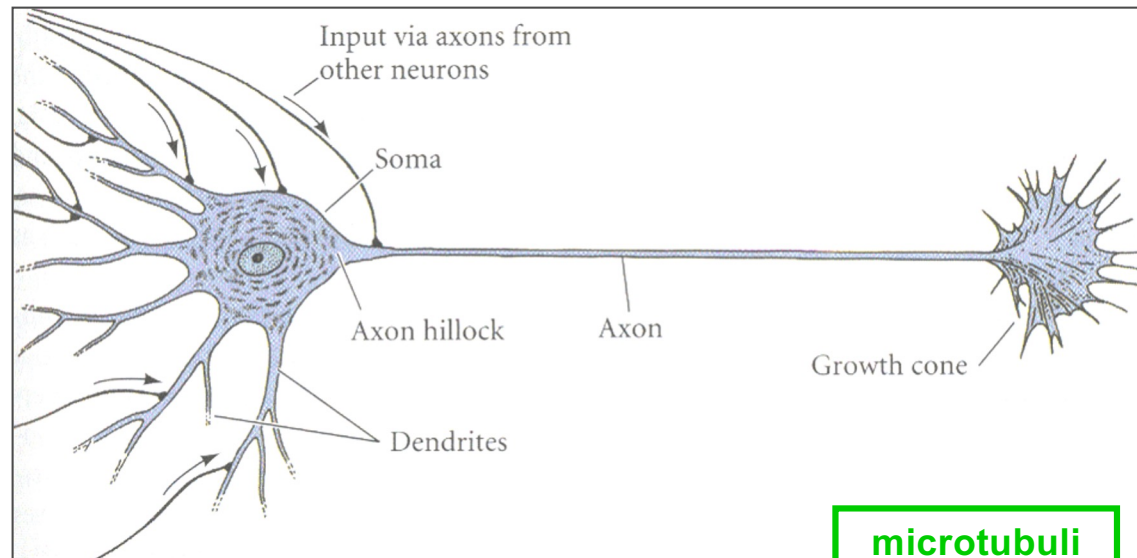


Struttura della cellula nervosa



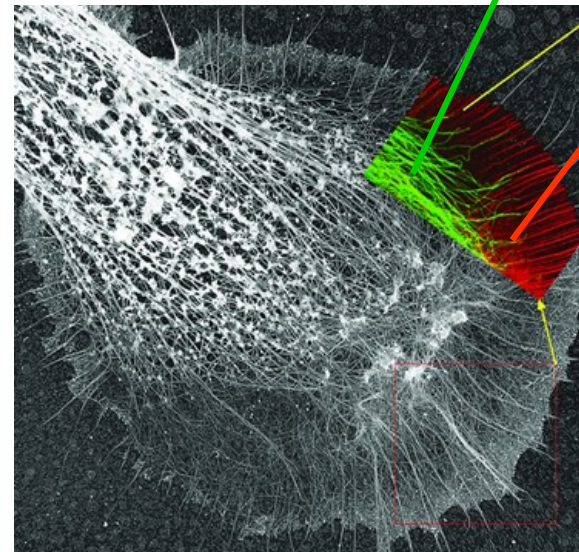
Rappresentazione schematica di una cellula nervosa
(sono rappresentate le principali strutture e le loro funzioni)

Il “cono di crescita” del neurone

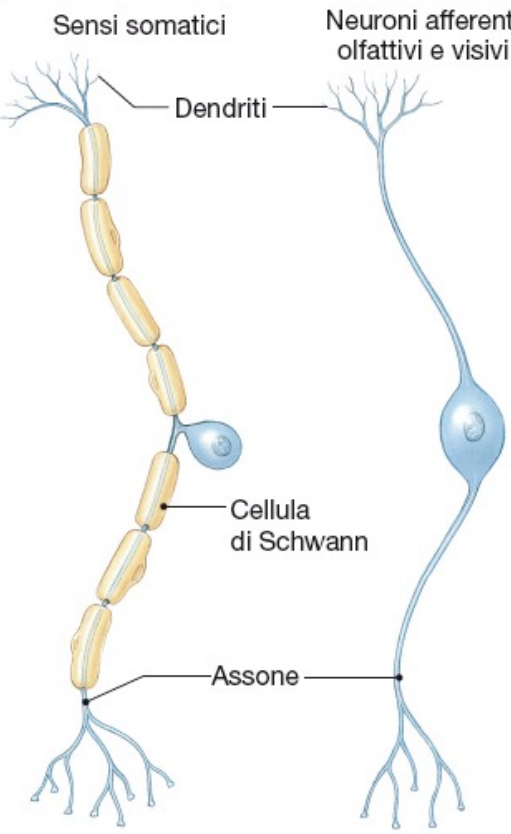
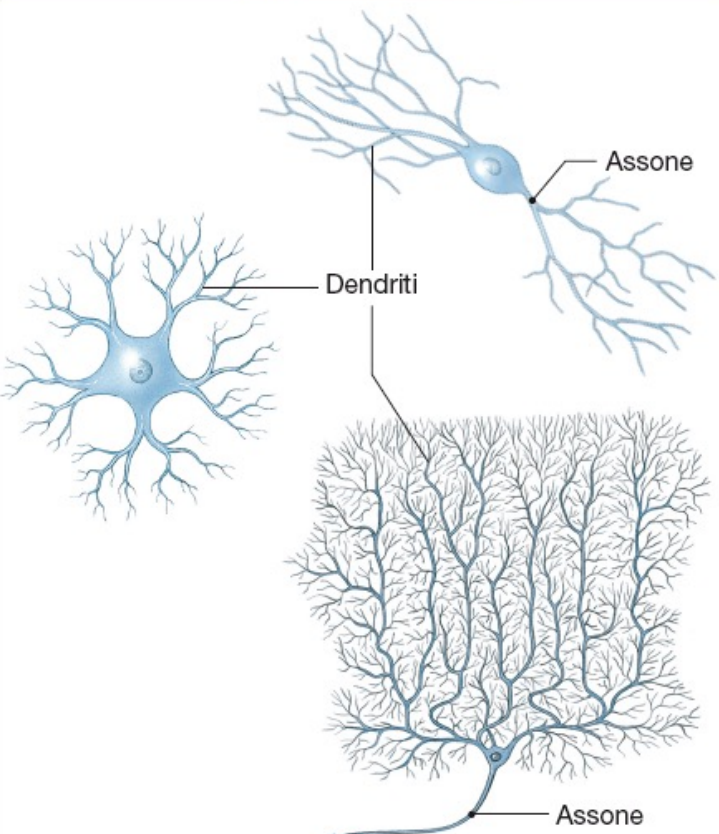
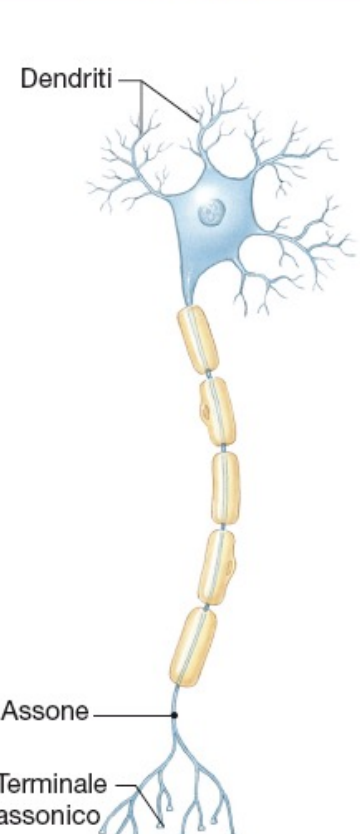


Santiago Ramon y Cajal, 1890:

“...il cono di crescita dell’assone è dotato di rapidi movimenti e di una certa forza motrice che gli conferisce la capacità di muoversi in avanti e di superare ostacoli...”



Esistono diversi tipi di neuroni

Neuroni sensoriali		Interneuroni del SNC		Neuroni efferenti
Sensi somatici Neuroni afferenti olfattivi e visivi  Pseudounipolare Bipolare		 Anassonico Multipolare		

(a) I neuroni pseudounipolari hanno un singolo processo chiamato assone. Durante lo sviluppo, il dendrite si è fuso con l'assone.

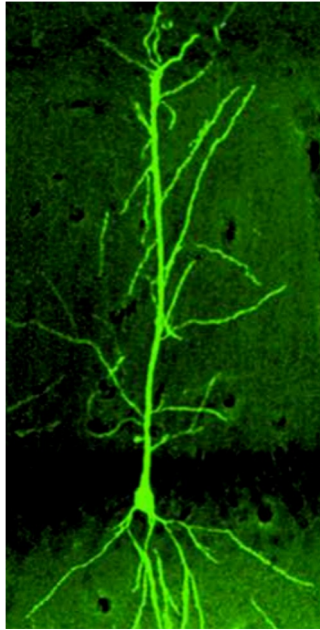
(b) I neuroni bipolari hanno due processi più o meno uguali che si dipartono dal corpo cellulare posto al centro.

(c) Gli interneuroni anassonici del SNC non hanno assoni evidenti.

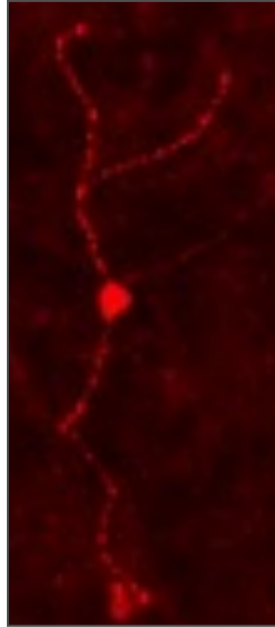
(d) Gli interneuroni multipolari del SNC sono molto ramificati, ma non hanno processi lunghi.

(e) Un neurone efferente multipolare tipico ha dai 5 ai 7 dendriti, ognuno dei quali si ramifica da quattro a sei volte. Ogni singolo, lungo assone si può ramificare più volte e finisce con delle dilatazioni, i terminali assonici.

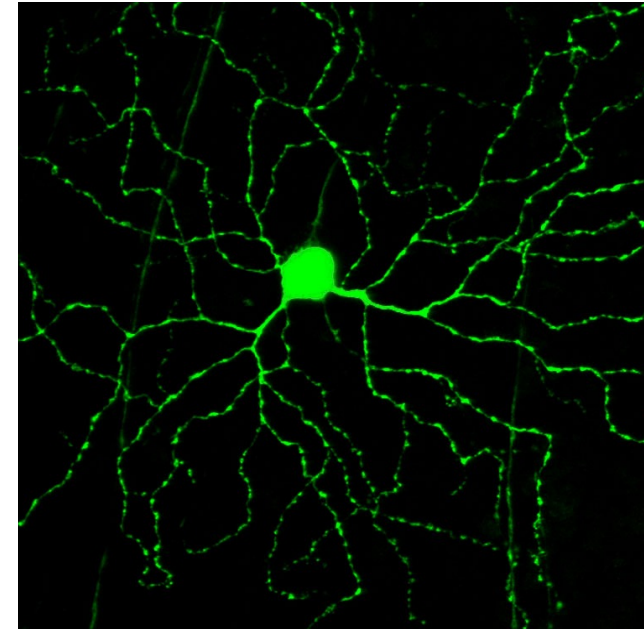
Esistono diversi tipi di neuroni



*neurone piramidale
(corteccia cerebrale)*



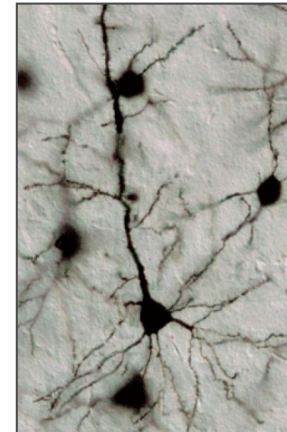
*interneurone
(corteccia cerebrale)*



cellula gangliare (retina)

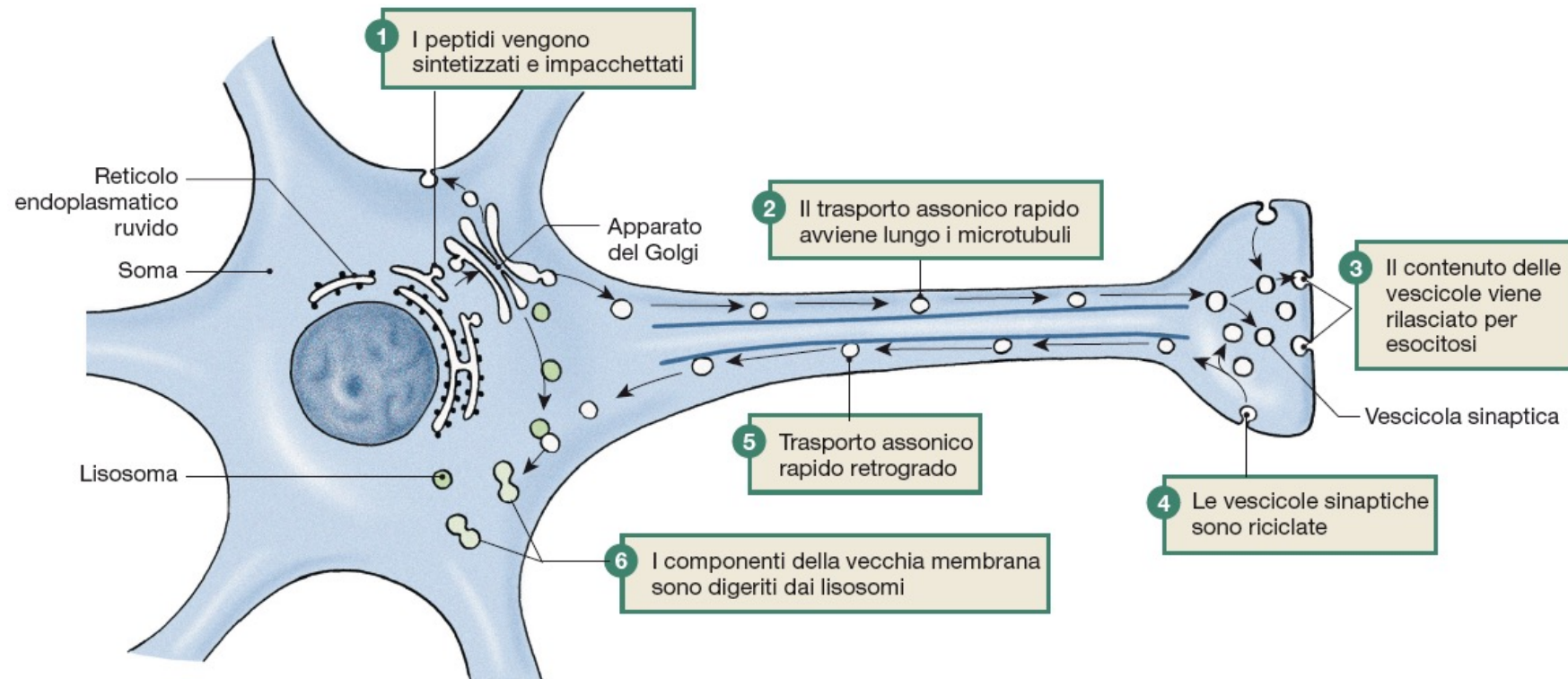


*cellula di Purkinjie
(cervelletto)*



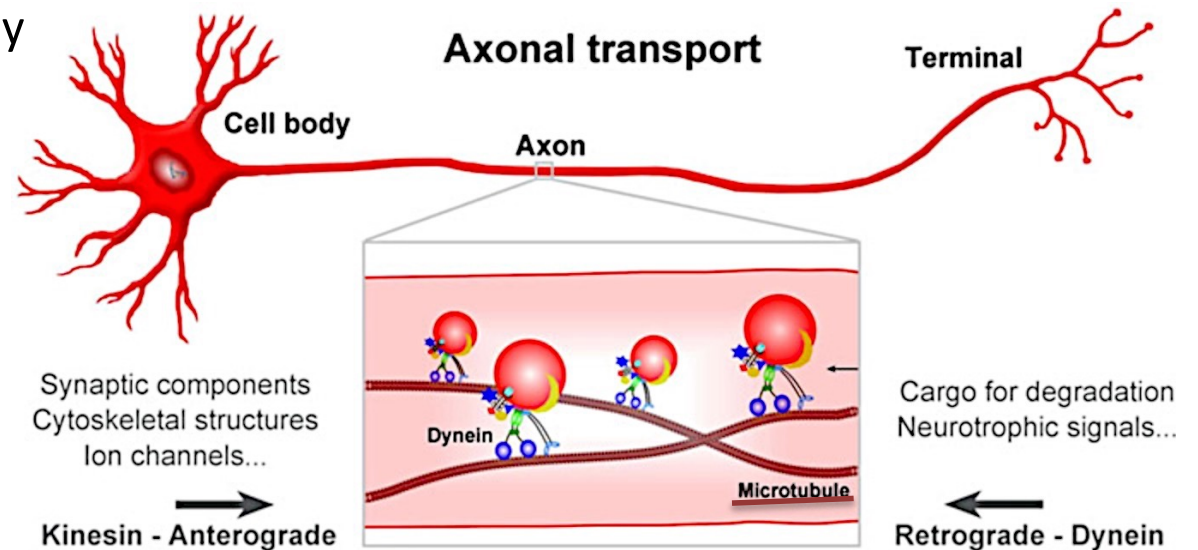
*neurone piramidale
e interneuroni
(corteccia cerebrale)*

Struttura della cellula nervosa – trasporto assonico

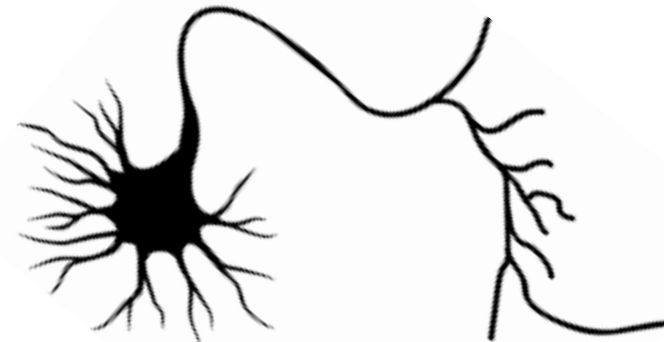


Transmission along the nerve fiber/axon

- Axons may be more than 1 m long
 - Anterograde and retrograde **intracellular transport** along axon
- Relatively slow <40cm/day



- **Action Potential conduction** → Fast (up to ~60 m/s)



Electrical properties

■ Membrane potential

- ☐ Difference in charge inside and outside the cell
- ☐ Plasma membrane barrier separating charges
- ☐ Ion concentrations differ between the inside and outside of the cell
- ☐ Polarized

■ Resting membrane potential

- ☐ When neurons are not sending signals

Basi ioniche del potenziale di membrana a riposo

La membrana è isolante e selettivamente permeabile

Proteine trasportatrici e canali di membrana generano un gradiente

elettrico – separazione di cariche

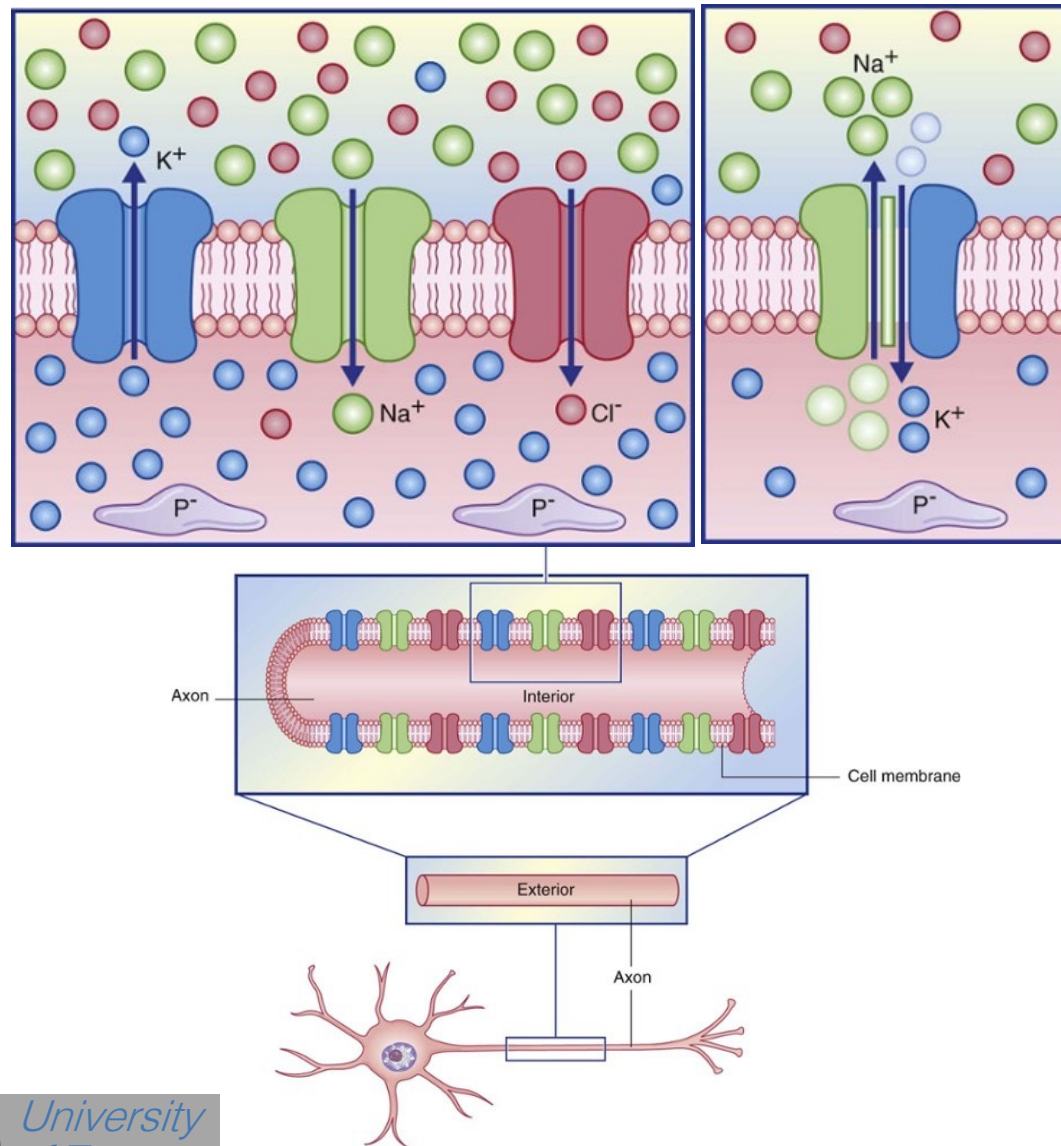
chimico – differenze di concentrazione

Tra i due lati di una membrana esiste un **gradiente *elettrochimico***

Le cellule restano invece in equilibrio osmotico perché l' H_2O diffonde liberamente

Il **gradiente elettrico** tra il liquido extra- ed intracellulare **si definisce come (*differenza di*)
potenziale di membrana (*a riposo*)**

Differences in ionic concentration across the resting neuronal membrane



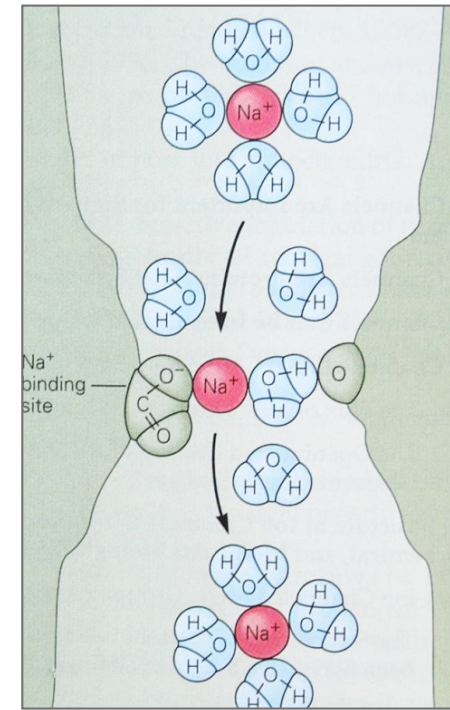
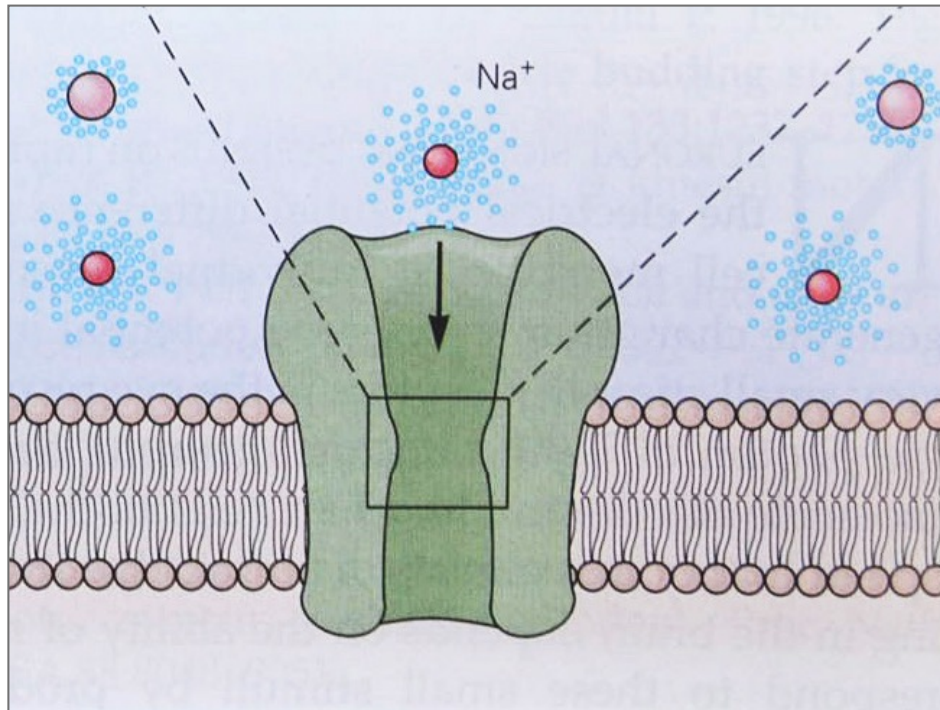
- **Passive (non-gated) ion channels** are open at all times, permitting ions to move across the membrane.
- **Channel pumps** are energy-driven ion exporters and/or importers designed to maintain steady-state ion concentrations. The **Na⁺-K⁺ exchange pump** (usually referred to as the *sodium pump*) is vital to maintenance of the resting membrane potential.

Canali ionici di membrana

Principali caratteristiche:

- *sono proteine transmembrana composte da molte subunità*
- *conducono ioni (fino a 10^8 ioni/s)*
- *il flusso di ioni è passivo (non richiede energia metabolica)*
- *il flusso di ioni dipende da un gradiente elettrochimico*
- *riconoscono e selezionano ioni specifici*
- *si aprono e si chiudono in risposta a stimoli elettrici, chimici e meccanici*
- *sito d'azione di droghe, farmaci, veleni e tossine*

Selettività dei canali ionici: “setaccio molecolare”



La mobilità degli ioni dipende dal diametro e dalla composizione chimica (aminoacidica) del poro che agisce come “setaccio molecolare”

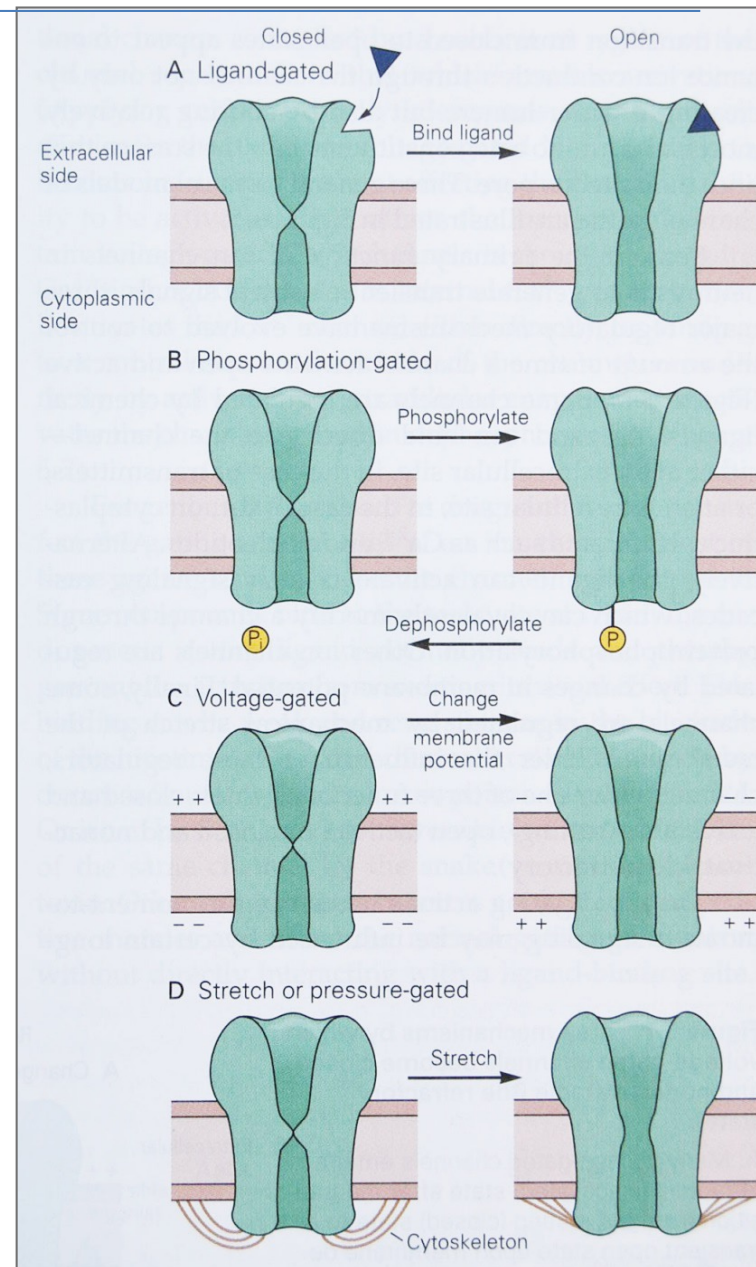
Canali ad apertura/chiusura controllata

Canali a controllo di ligando

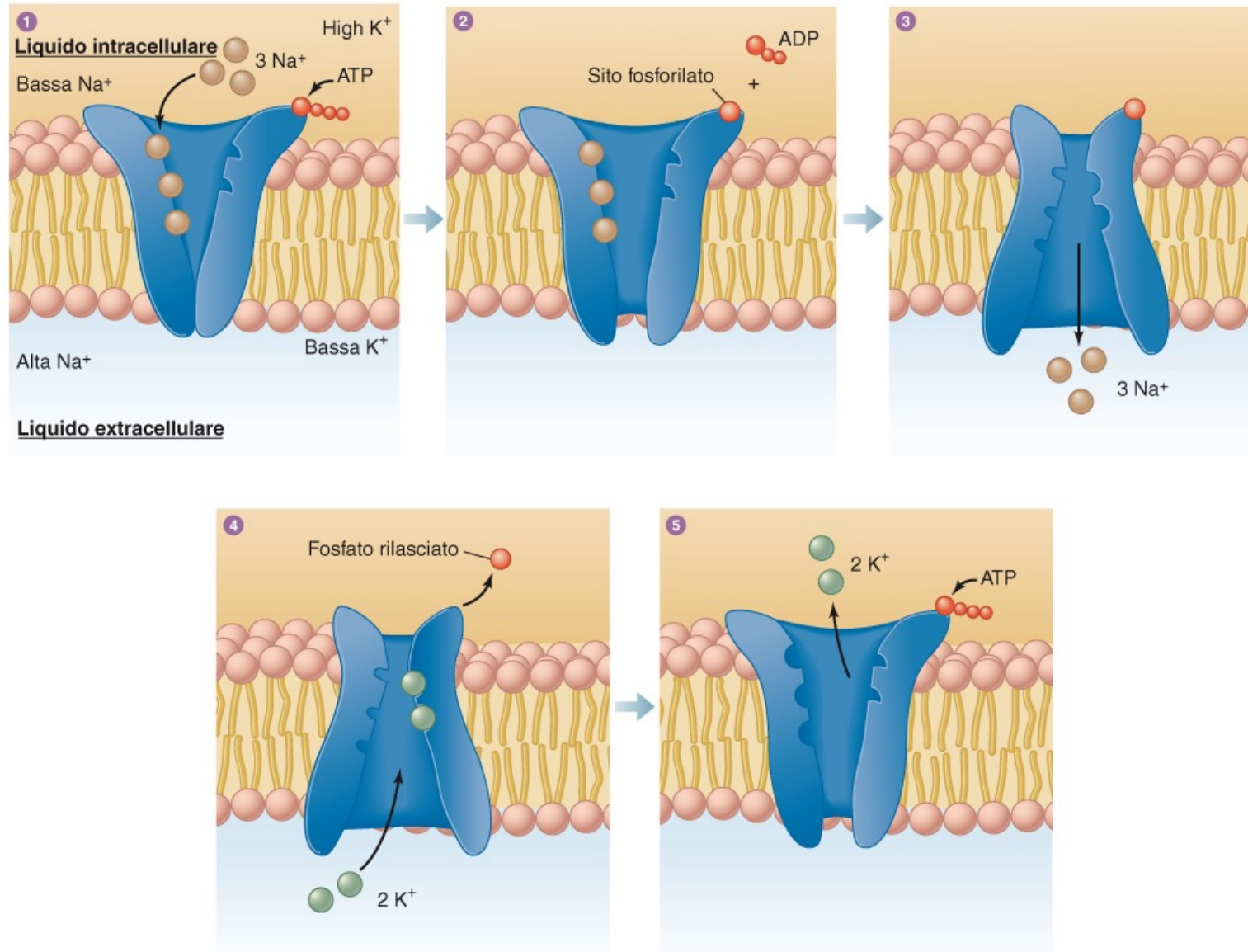
Canali a controllo di fosforilazione

Canali voltaggio-dipendenti

Canali regolati meccanicamente



Attività della $\text{Na}^+\text{-K}^+\text{-ATPasi}$



La $\text{Na}^+\text{-K}^+\text{-ATPasi}$ utilizza l'energia dell'ATP per pompare 3 Na^+ fuori dalla cellula e 2 K^+ dentro la cellula

Basi ioniche del potenziale di membrana a riposo

In TUTTI i tipi cellulari, la **membrana opera una separazione di cariche tra esterno e interno** della cellula

Na^+	<i>maggiormente concentrato</i>	all' esterno
Cl^-	<i>maggiormente concentrato</i>	all' esterno
K^+	<i>maggiormente concentrato</i>	all' interno
Anioni <i>(fosfati, proteine)</i>	<i>maggiormente concentrati</i>	all' interno
HCO_3^-	<i>maggiormente concentrato</i>	all' esterno
Ca^{2+}	<i>maggiormente concentrato</i>	all' esterno
Mg^{2+}	<i>variabile (in equilibrio o + concentrato all' interno)</i>	

In condizioni di riposo, l' interno della cellula è sempre più negativo dell' esterno

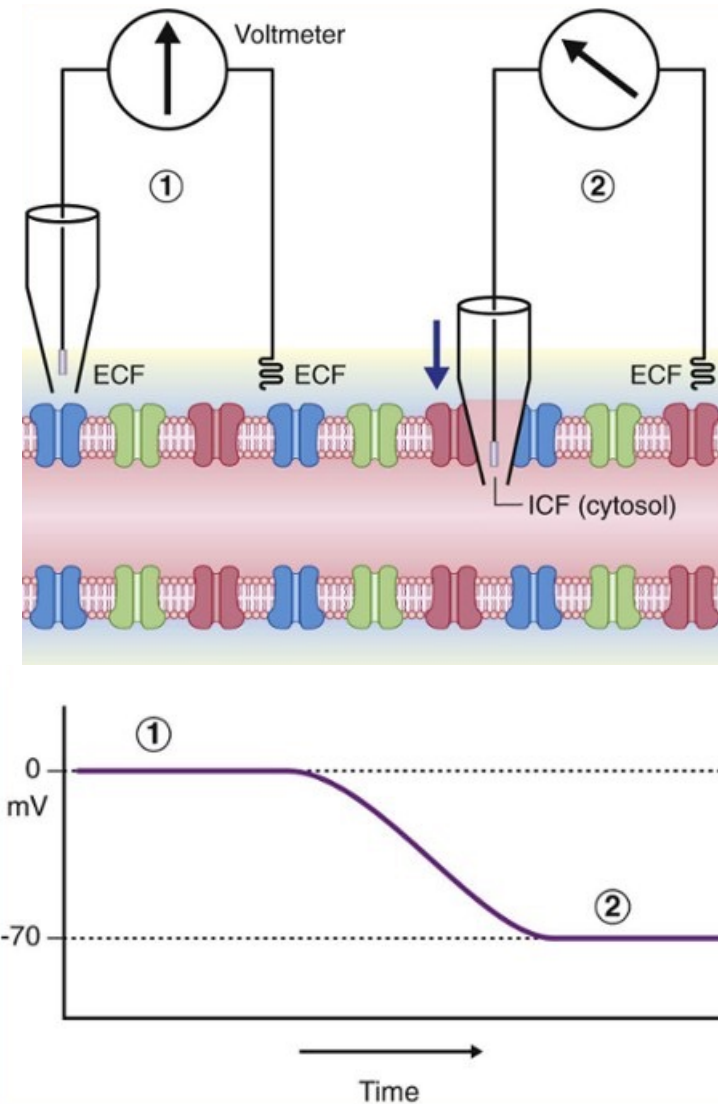
Resting Membrane Potential

The membrane potential of the resting (inactive) neuron is generated primarily **by differences in concentration of the sodium (Na^+) and potassium (K^+) ions** dissolved in the aqueous environments of extracellular fluid (ECF) and cytosol.

The equilibrium state exists when the **concentration gradient** is exactly **balanced by the voltage gradient**.

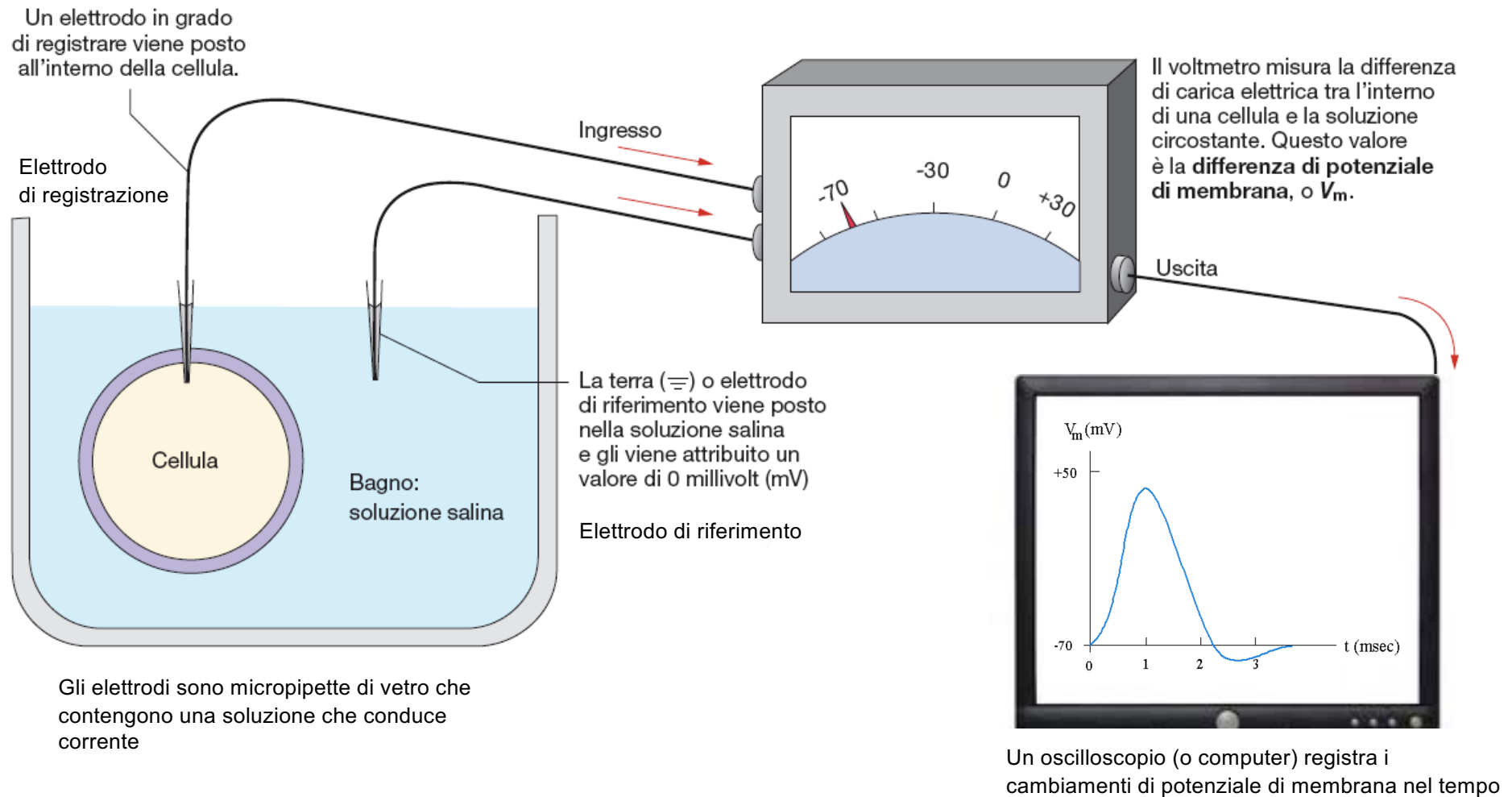
Ion	<u>Concentration (mmol/L)</u>		Equilibrium Potential (mV)
	Cytosol	Extracellular	
K^+	100	5	-90
Na^+	15	150	+60
Cl^-	13	50	-70

The sum of the fractions for K^+ and Na^+ yields an outcome of **-70 mV**. The effect of the chloride ion on resting potential is minimal, because its equilibrium potential is the same as the resting potential.

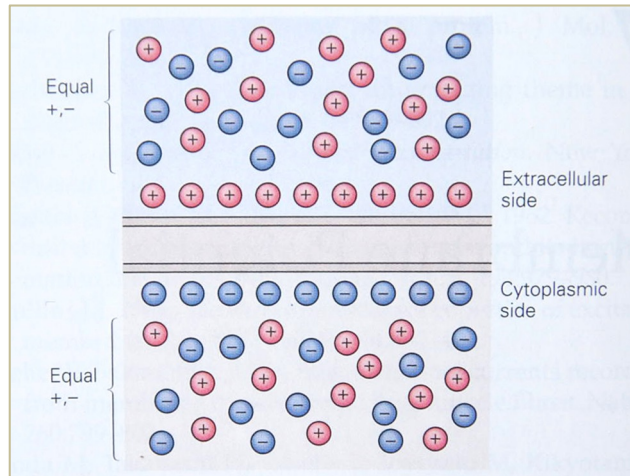


Registrazione del potenziale di membrana

Si misura sempre relativamente all' ambiente extracellulare



Basi ioniche del potenziale di membrana a riposo



Na^+ , Cl^-

K^+ ,
Anioni
(aminoacidi, proteine)

Si definisce potenziale di membrana a riposo la differenza di potenziale attraverso la membrana in assenza di segnale

Il potenziale di membrana a riposo è dovuto alla separazione di cariche ai due lati della membrana

L' eccesso di cariche + all' esterno e di cariche – all' interno rappresenta solo una minima frazione del numero totale di cariche

Il flusso di cariche attraverso la membrana è regolato da **due tipi di canali**:

canali a riposo – sempre aperti, mantengono il potenziale di membrana a riposo

canali regolati – normalmente chiusi a riposo

Alteration in the level of polarization of target neurons

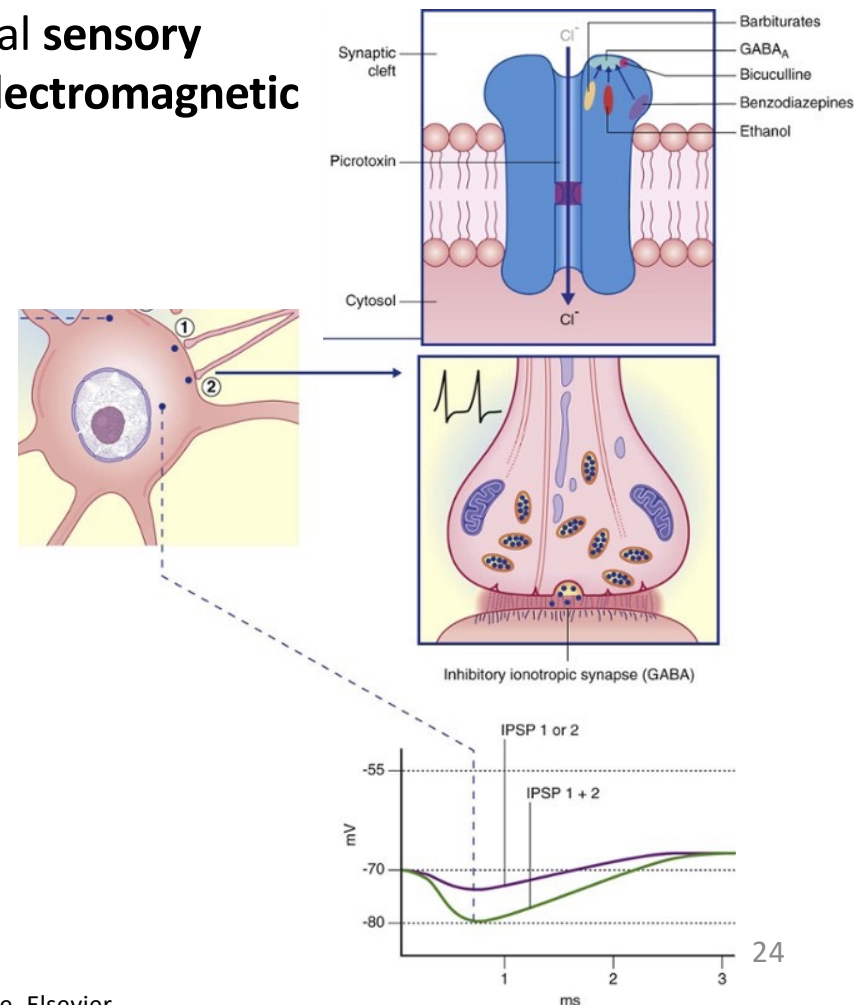
Transmitter-gated channels abound in **postsynaptic** membranes: are activated **directly or indirectly** by **transmitter molecules**

Transduction channels are activated by peripheral **sensory** stimulation: for example **mechanical, thermal, electromagnetic** (in the retina), etc etc.

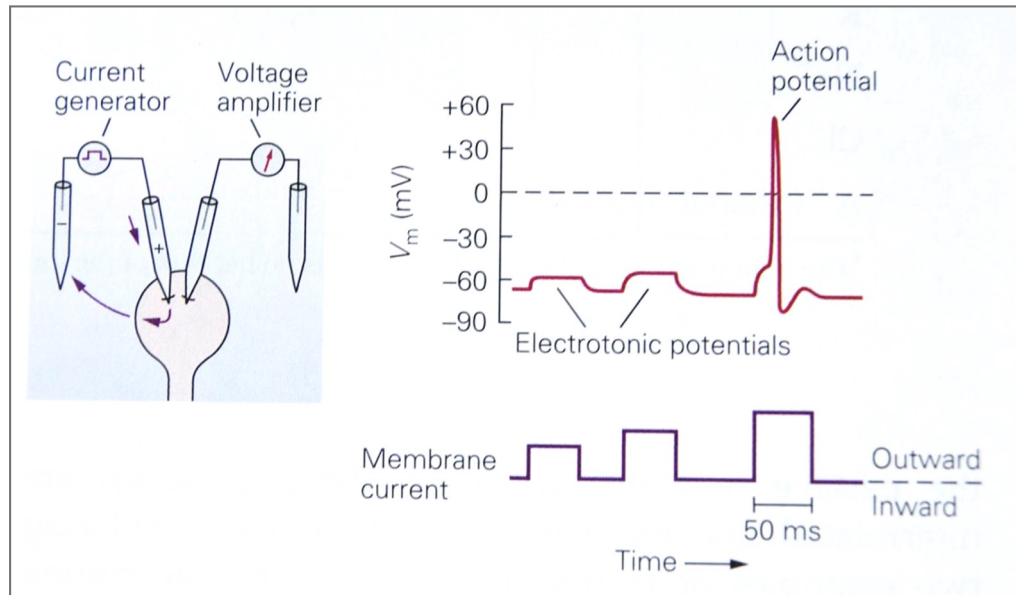
Polarization may **positively or negative** altered:

- **Depolarizing** neurotransmitters/stimuli **decrease membrane potential** → **excitatory**
- **Hyperpolarizing** transmitters (i.e. GABA) **may increase the resting potential** → **inhibition**

Hyperpolarization

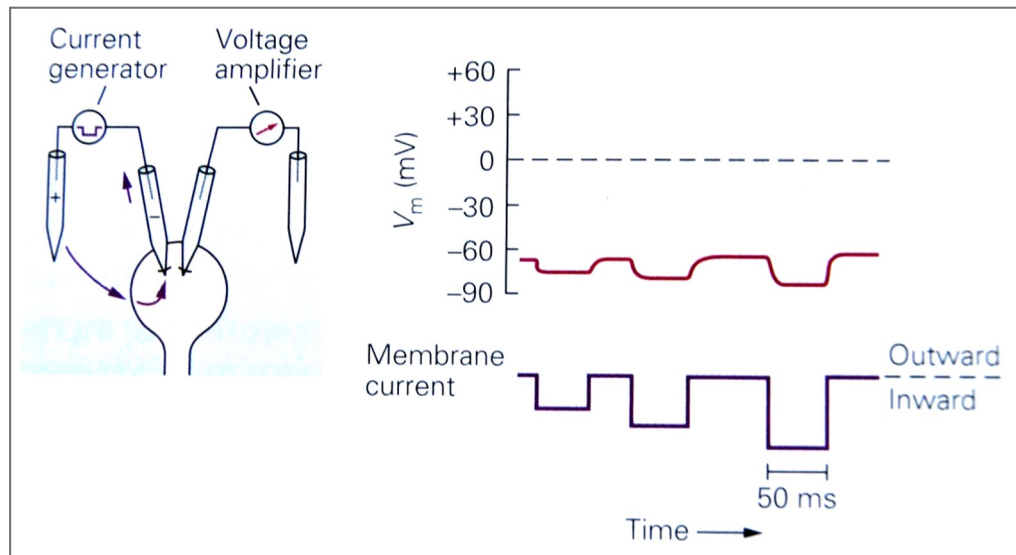


Depolarizzazione e iperpolarizzazione



Depolarizzazione: riduzione della separazione di carica (= potenziale meno negativo)

Quando la depolarizzazione supera una soglia, si aprono i canali voltaggio-dipendenti e si genera il *potenziale d'azione*



Iperpolarizzazione: aumento della separazione di carica (= potenziale più negativo)

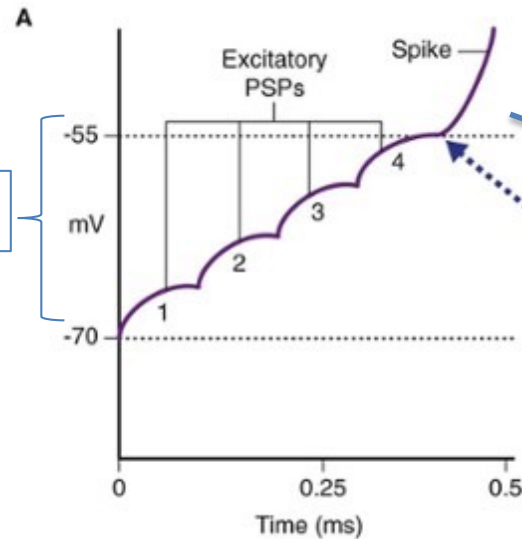
Potenziali elettrotonici: piccole variazioni (passive) del potenziale di membrana, che non determinano apertura di canali regolati. Le risposte iperpolarizzanti sono sempre elettrotoniche

Induction of Action Potential

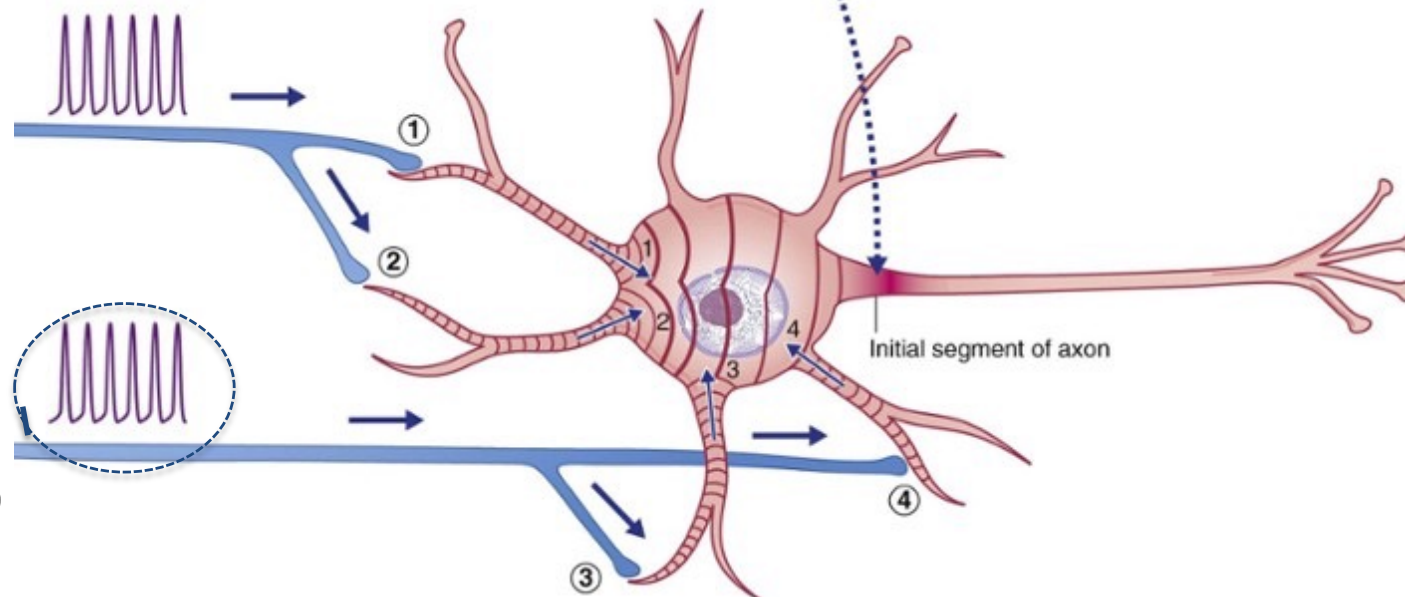
Graduated potentials

Summation of multiple postsynaptic depolarization potentials (1+2+3+4)

Temporal Summation (multiple spikes)



When depolarization reach a threshold level **Voltage-gated channels open**, initiating the **Action Potential in the axon**



Features of the Action Potential

Action Potential in the axon is an “all or none” response

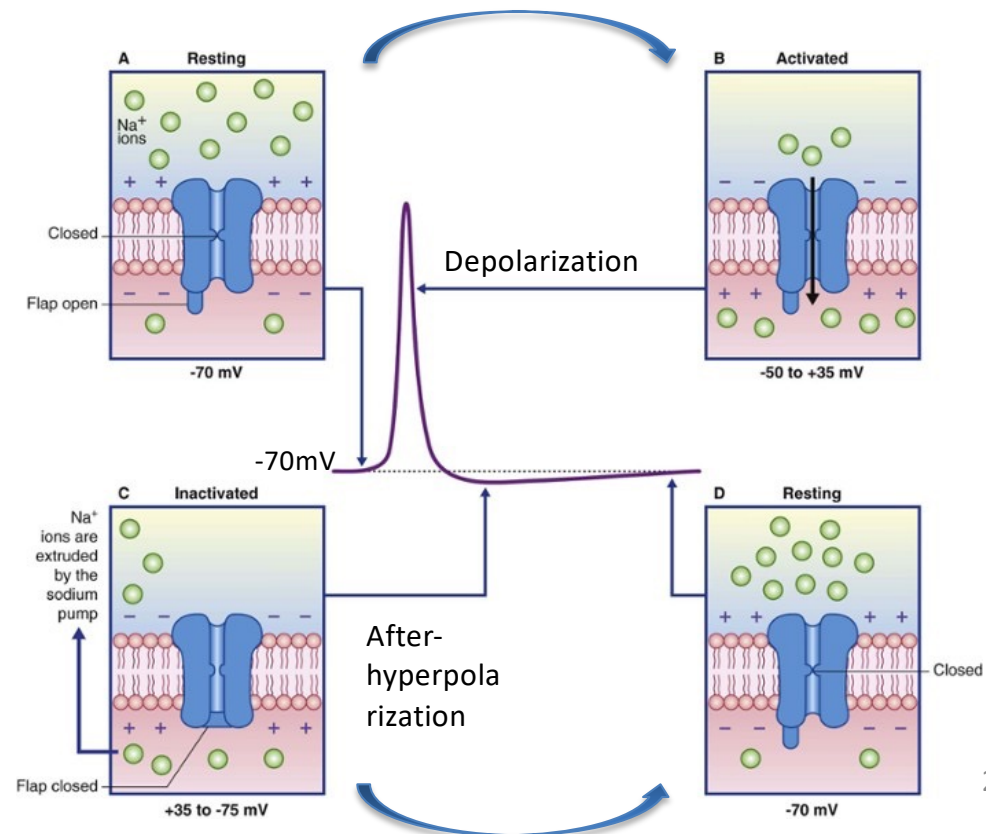
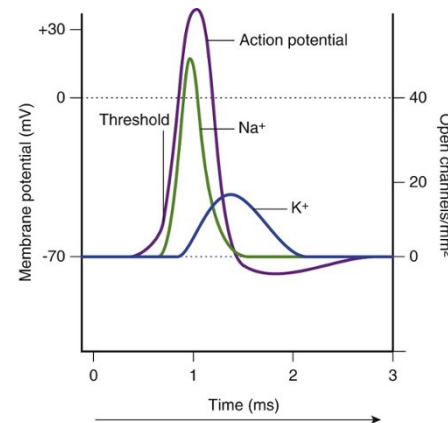
The action potential mainly depends on **Na⁺ inflow** and **subsequent K⁺ outflow**

The phases of the Action Potential are determined by the characteristics of the Ion Channels (kinetics of opening, inactivation)

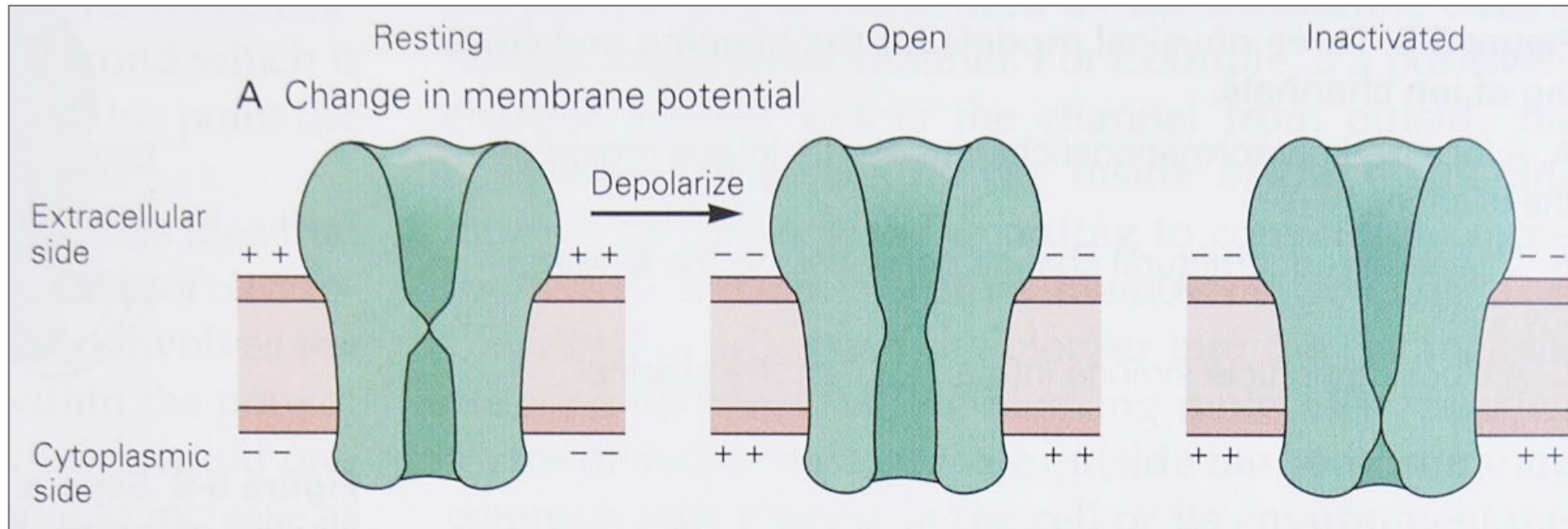
Sodium channels are playing a major role

Jove videos:

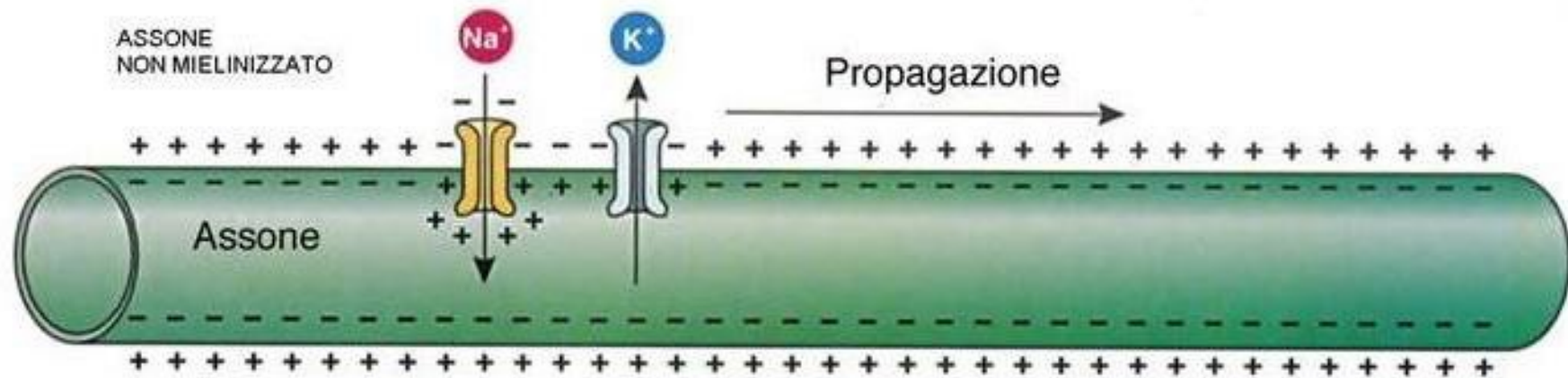
<https://app.jove.com/it/science-education/v/14893/action-potential-phases-of-stimulation>



Chiusura (inattivazione) dei canali voltaggio-dipendenti



Propagazione del potenziale di azione

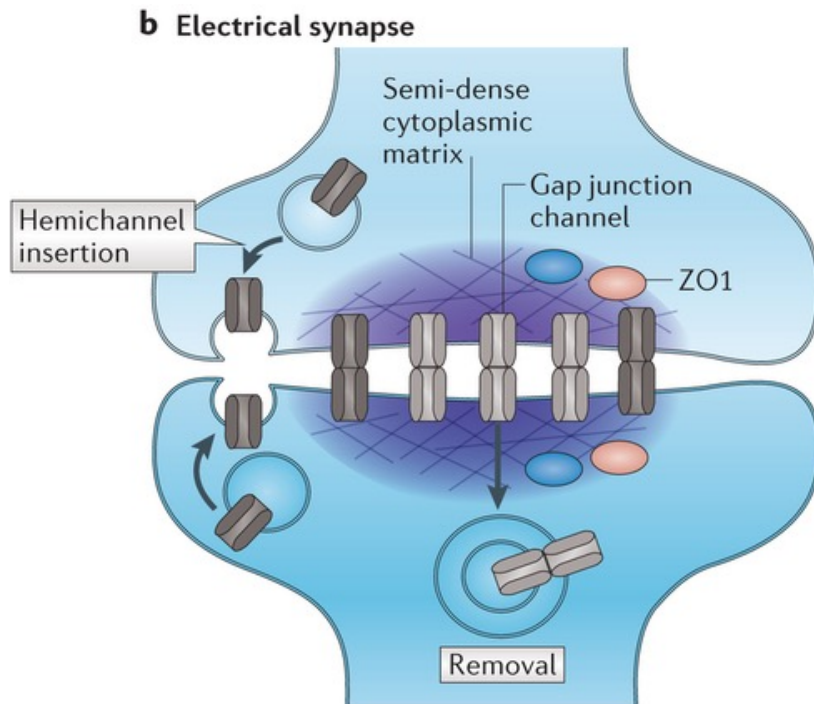


<https://app.jove.com/it/science-education/v/14894/propagation-of-action-potentials>

Neurons communicate through Synapses

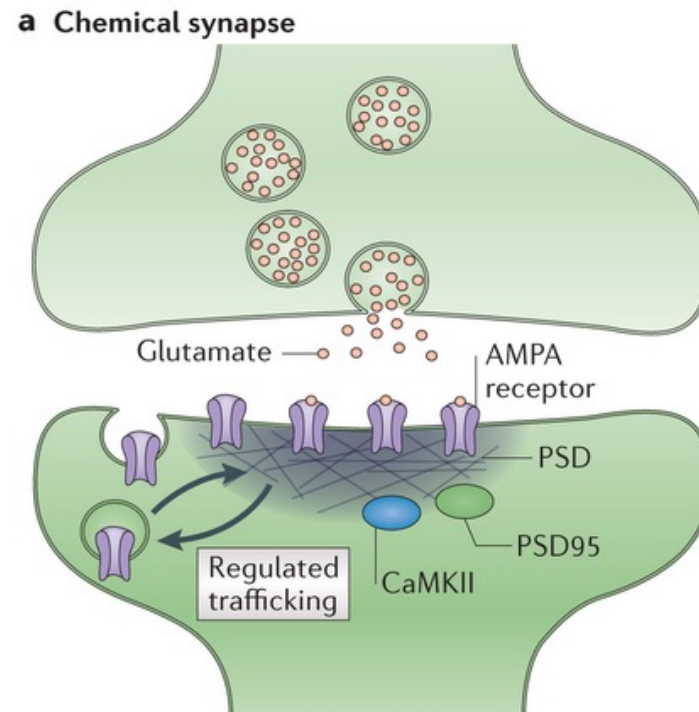
Electrical Synapses

Gap Junctions between contiguous neurons. **Cytoplasmic continuity.**
Rare in mammals.



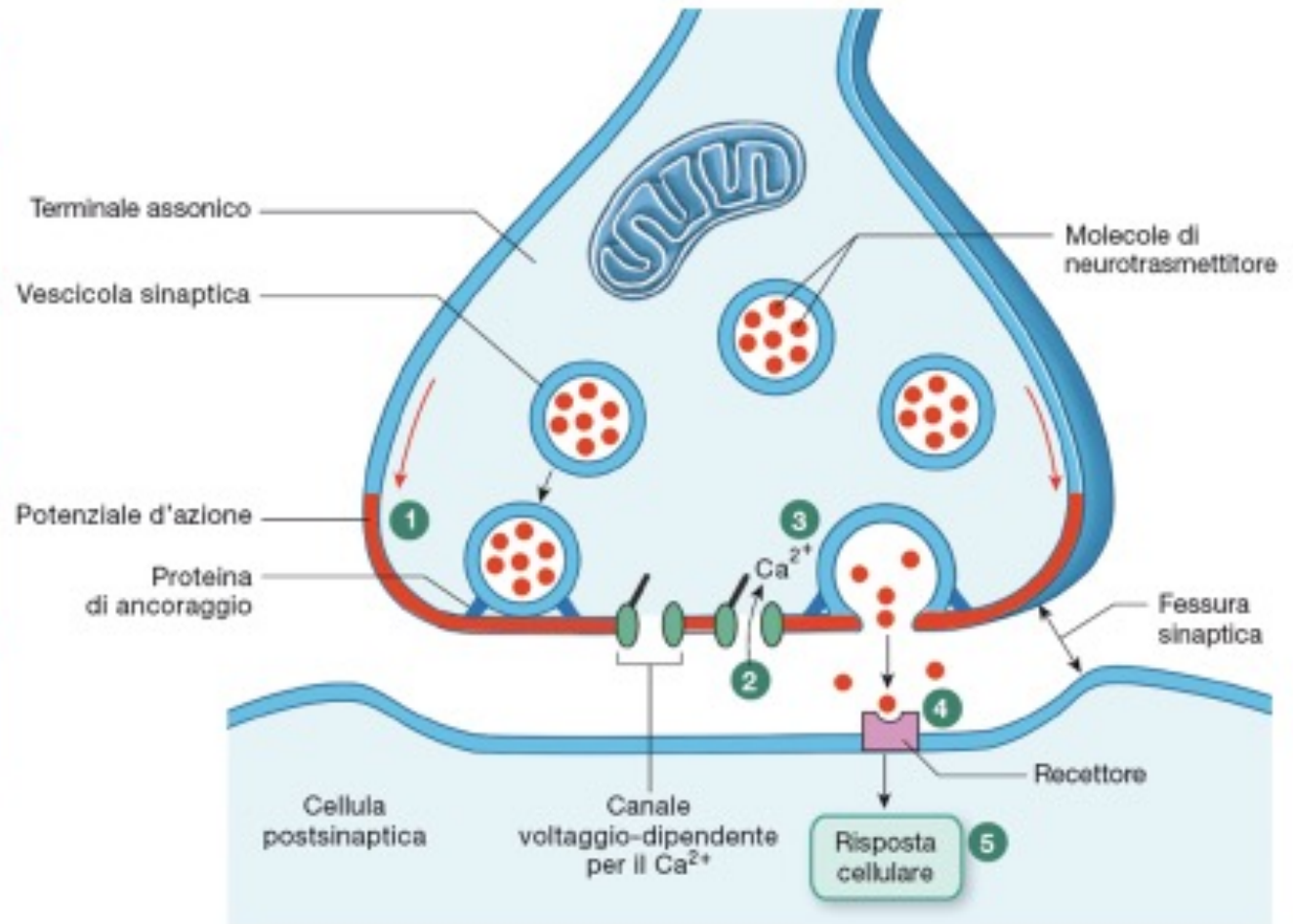
Chemical Synapses

Depend on the release of a **transmitter** substance.
Conventional Synapses in mammalian NS.



Trasmissione sinaptica

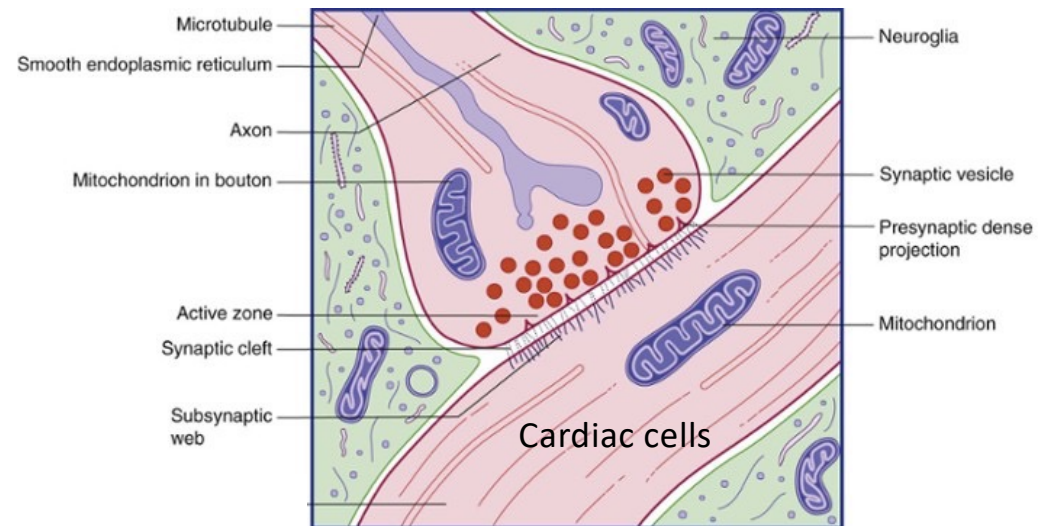
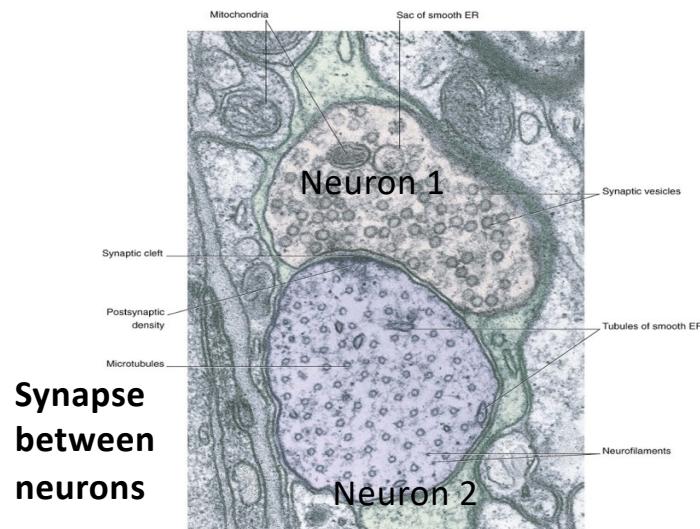
- 1 Un potenziale d'azione depolarizza un terminale assonico.
- 2 La depolarizzazione apre i canali per il Ca^{2+} voltaggio-dipendenti e il Ca^{2+} entra nella cellula.
- 3 L'ingresso del calcio provoca l'esocitosi del contenuto della vescicola sinaptica.
- 4 Il neurotrasmettitore diffonde attraverso la fessura sinaptica e si lega ai recettori sulla cellula postsinaptica.
- 5 Il legame del neurotrasmettitore con i recettori avvia una risposta nella cellula postsinaptica.



Neurons communicate through Synapses

- With the arrival of action potentials, **voltage-gated calcium channels** located in the **pre-synaptic** membrane are **opened**

In the case of small synaptic vesicles such as those containing **aminoacids** glutamate or γ -aminobutyric acid (GABA), **single spikes** are sufficient to yield some transmitter release. In the case of peptidergic neurons (releasing **neuropeptides**), **impulse frequencies of 10 Hz** or more are required



- Calcium channels** (directly or indirectly activated by neurotransmitters) located in the **post-synaptic** membrane are also **crucial to initiate responses** in the **post-synaptic cells** such as **neurons** or **other target** cells (for example cardiomyocytes in adrenergic response)

Neurotransmitters

Examples of Transmitters/modulators

Amino acids

- Glutamate
- γ -Aminobutyric acid (GABA)
- Glycine

Biogenic amines

- Acetylcholine
- Monoamines
- Catecholamines (dopamine, noradrenaline, adrenaline)
- Serotonin (5-HT)
- Histamine

Neuropeptides

- Vasoactive intestinal polypeptide (VIP)
- Substance P
- Enkephalin
- Endorphins
- Other

Adenosine

Nitric oxide (gaseous)

Specificity of action: receptors on target cell

Examples:

• Acetylcholine receptors

Nicotinic **Excitatory**-It is a multi-subunit Cation (Ca, Na, K) channel that opens

Muscarinic M1, M3, M5 **Excitatory**-Opens calcium channels

Muscarinic M2, M4 Mainly presynaptic-**Inhibitory**.

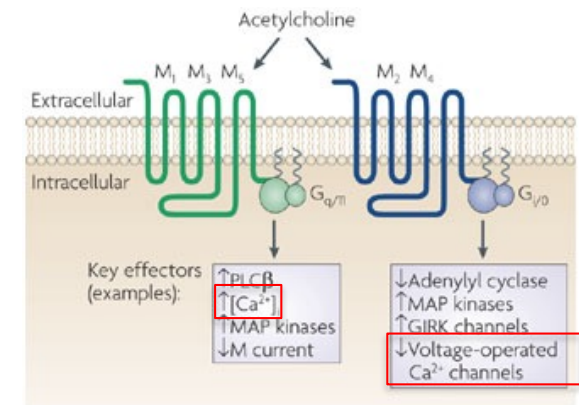
• Noradrenergic receptors

α_{1A-B-D} *Postsynaptic*—Phosphoinositol- **Excitatory**-Opens calcium channels

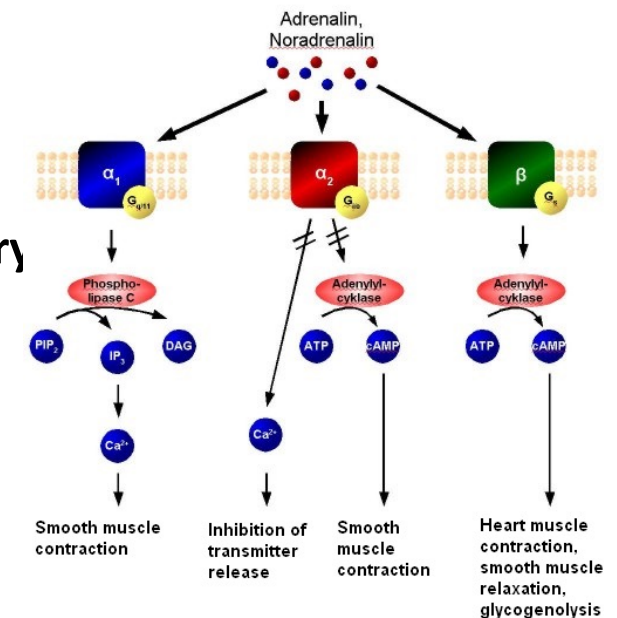
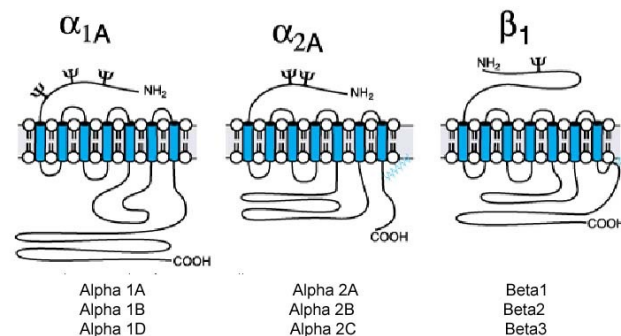
α_{2A-B-C} Mainly *presynaptic* -G protein direct-**Inhibitory**

β_1 *Postsynaptic*--Adenylate cyclase-**Excitatory**-Opens calcium channels

β_{2-3} *Post- and pre-synaptic*--Adenylate cyclase-**Inhibitory**



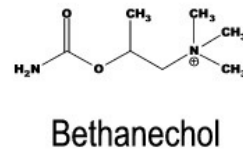
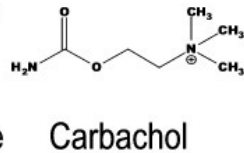
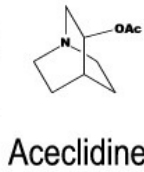
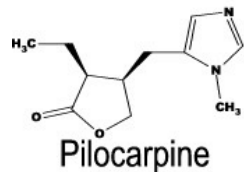
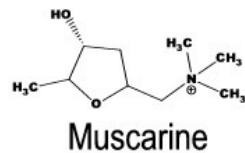
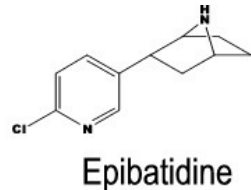
Nature Reviews | Drug Discovery



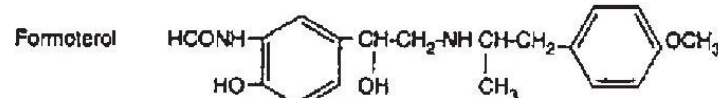
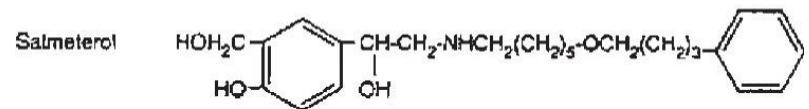
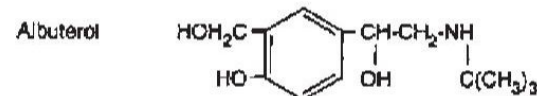
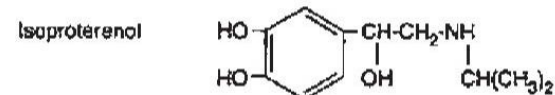
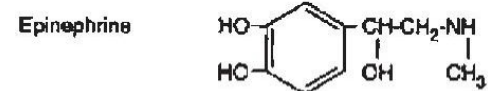
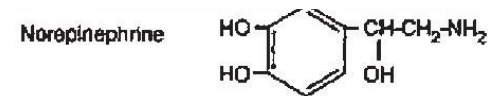
Receptor-specific Agonists and Antagonists

A few **examples of agonists**:

cholinergic receptor agonists



adrenergic agonists



All β

β_2

→therapeutically relevant
→important in research

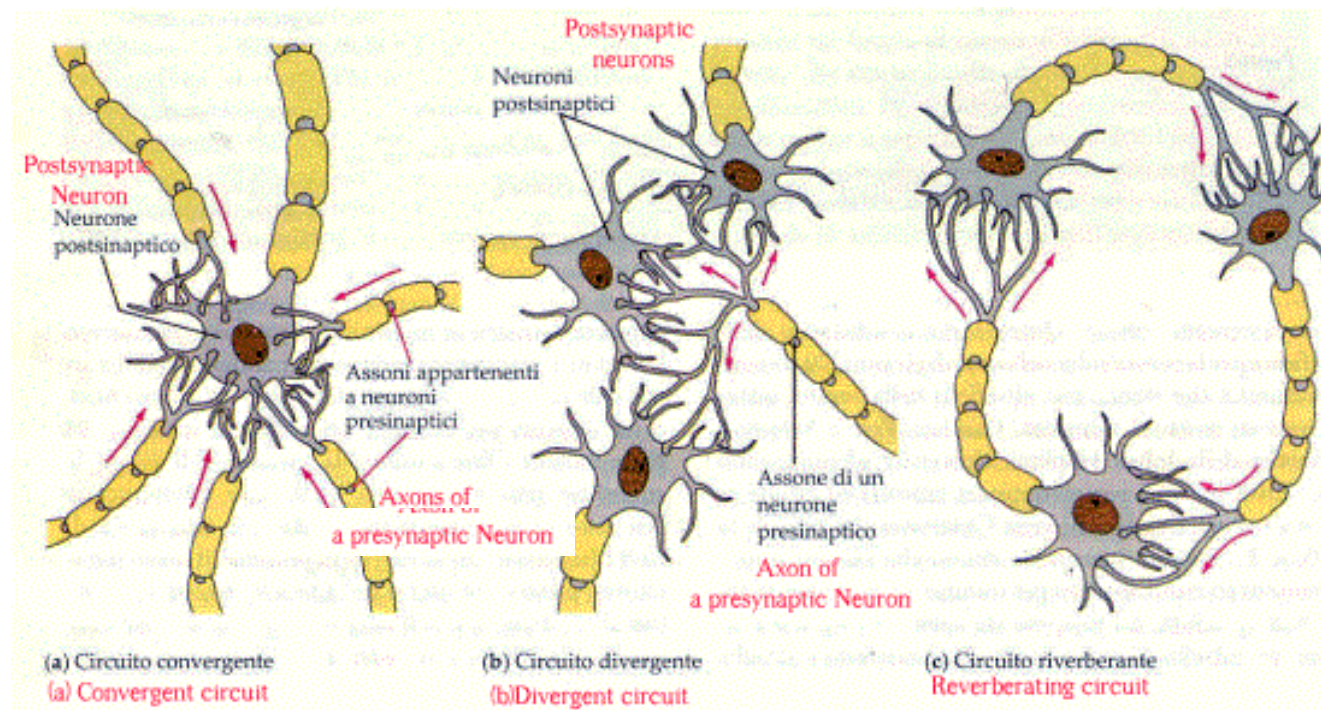
Neuronal Connections

Position related to Synapses

Pre-synaptic neurons

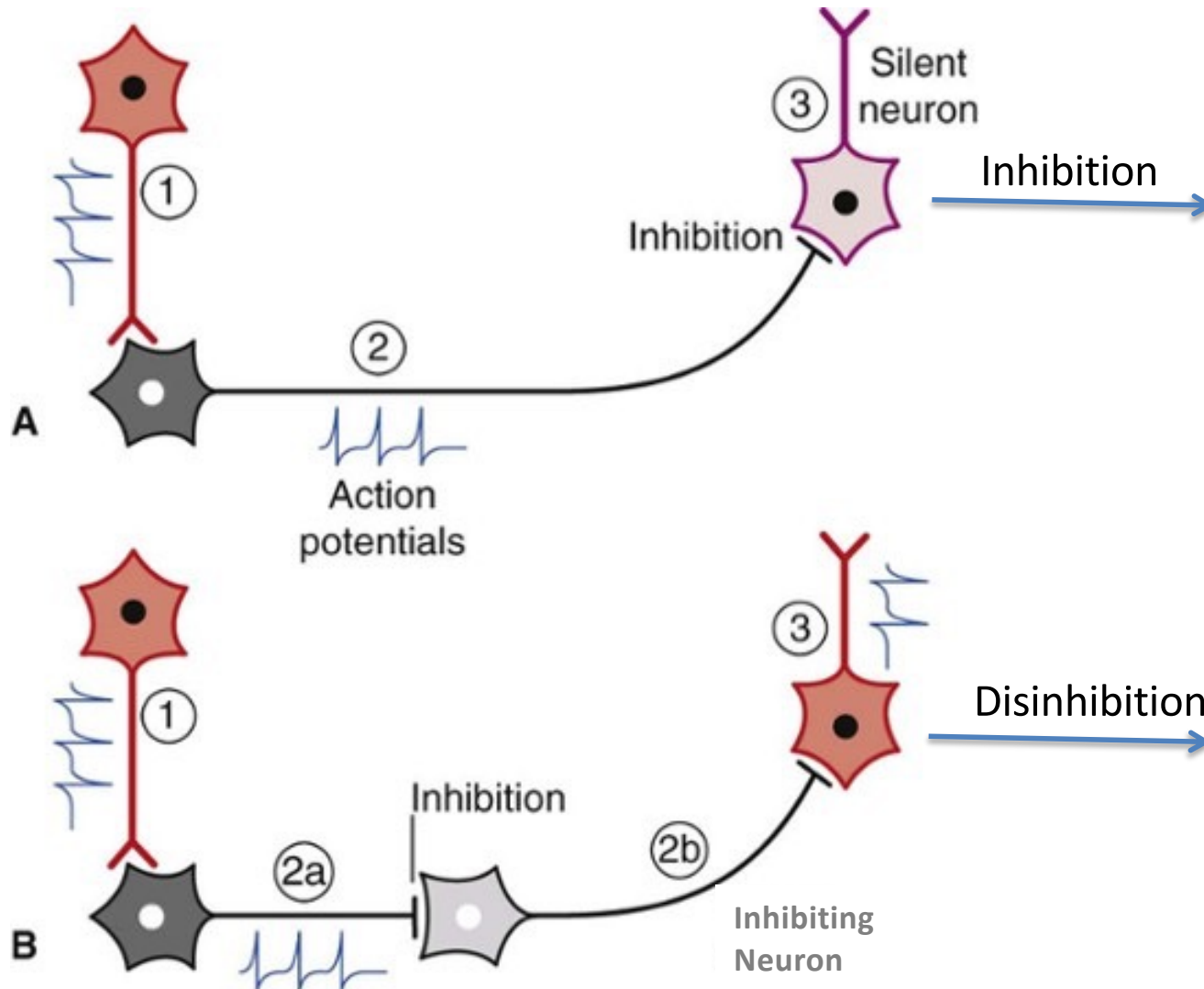
Post-synaptic neurons

Circuiti convergenti, divergenti e riverberanti:

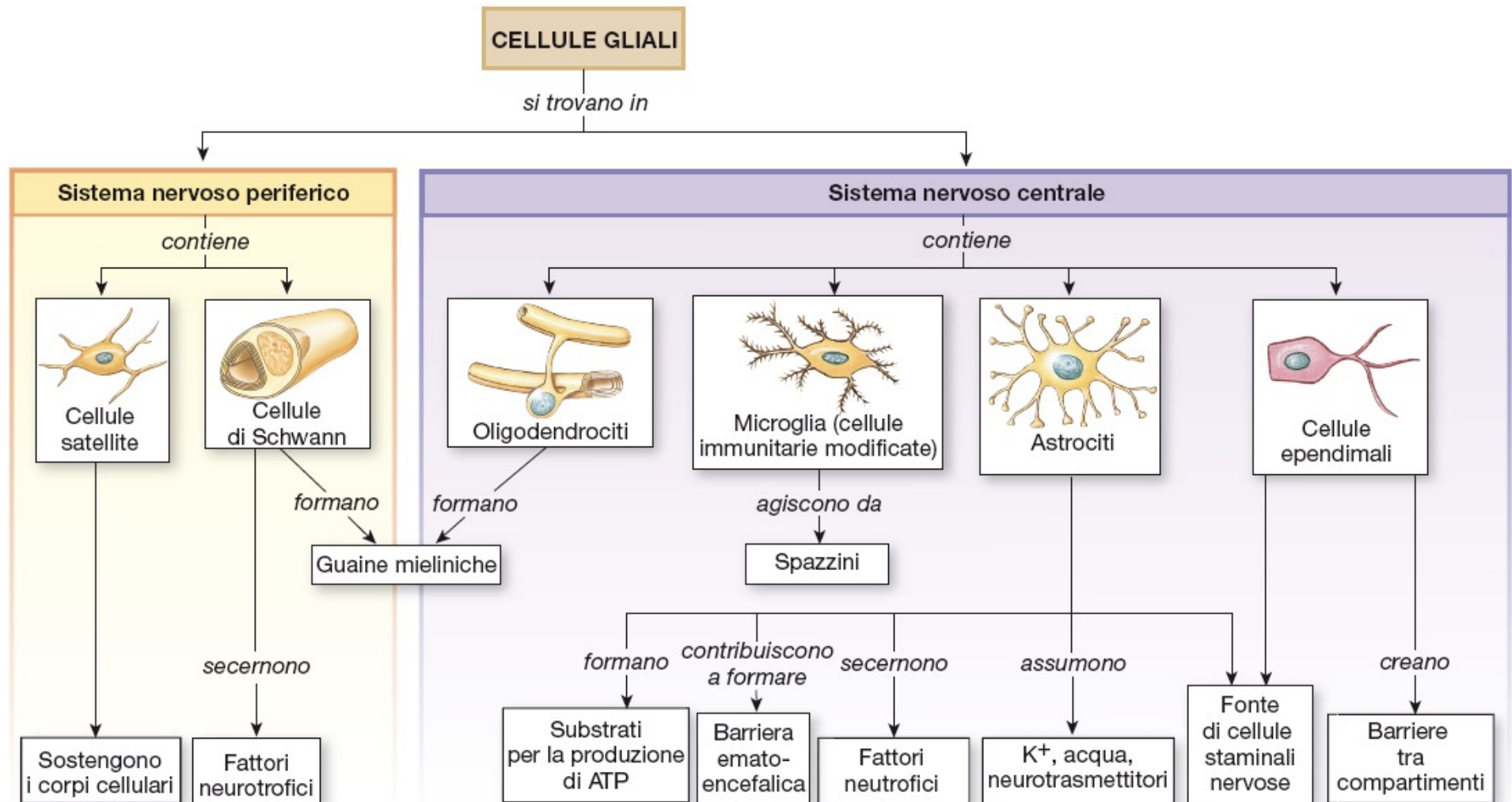


<https://app.jove.com/it/science-education/v/14900/neural-circuits>

Interneuron circuits



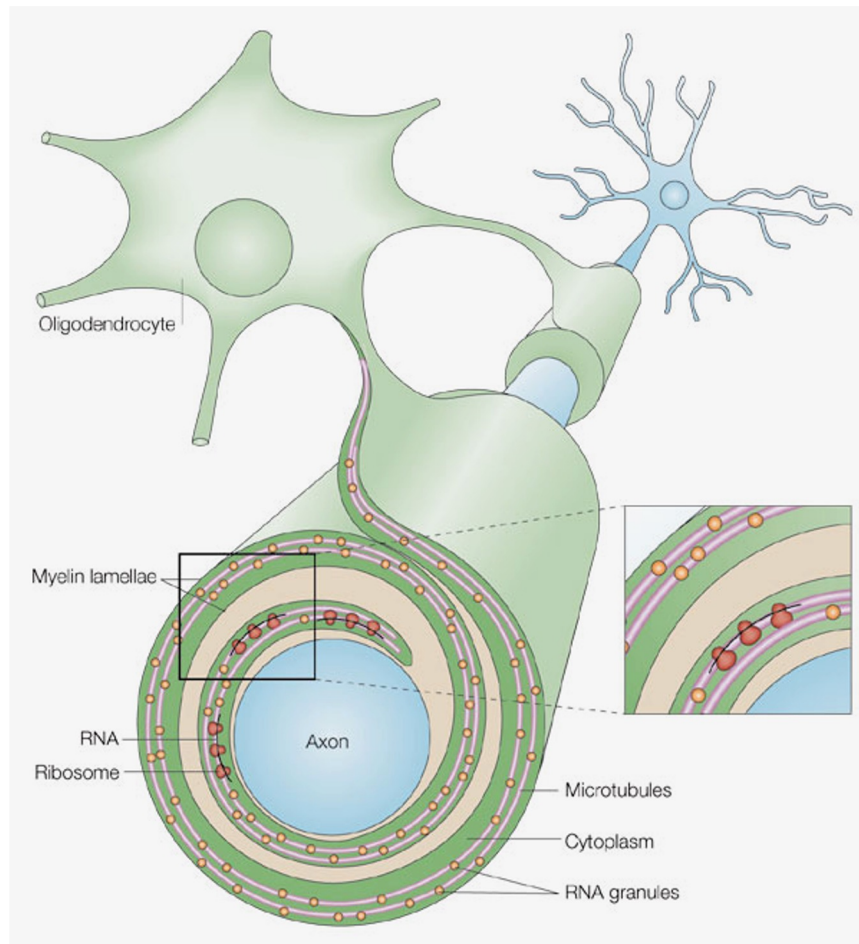
Struttura e funzione delle cellule gliali



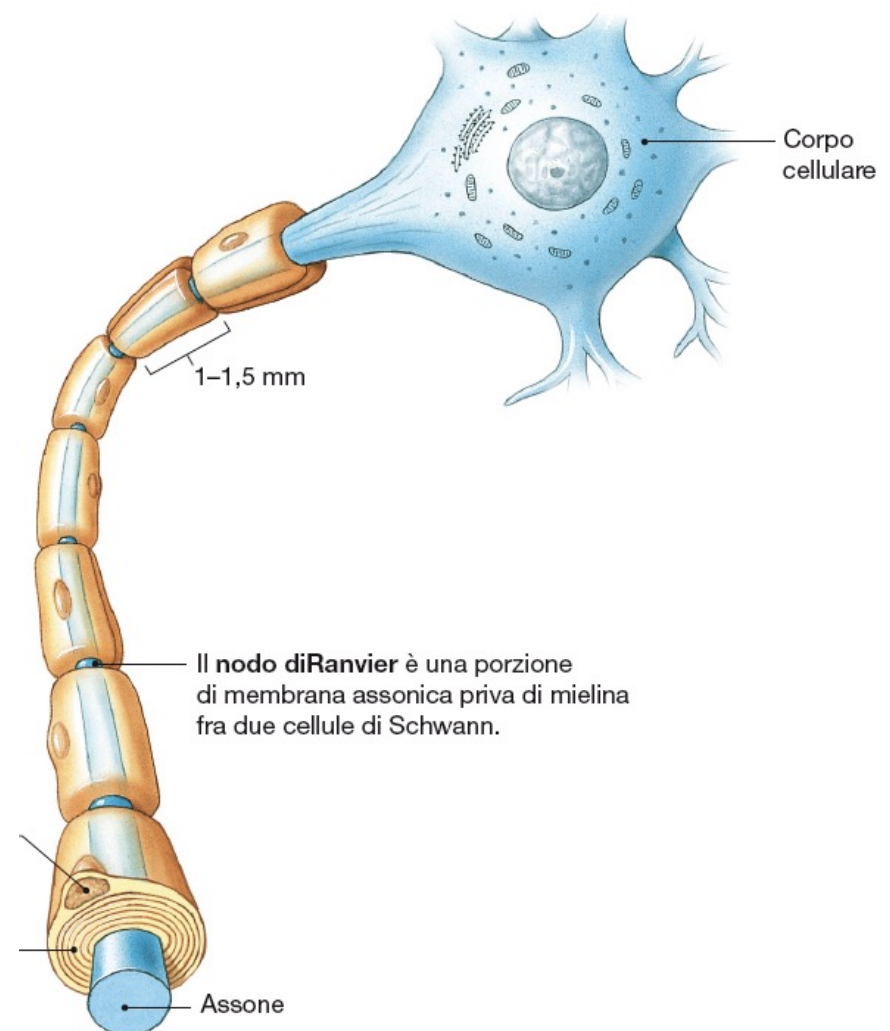
(b) Le cellule gliali e le loro funzioni

Jove video: <https://app.jove.com/it/embed/player?id=14996&language=it&t=1&s=1&fpv=1>

Le cellule gliali possono formare guaine di mielina



Gli **oligodendrociti** formano la mielina dei neuroni centrali

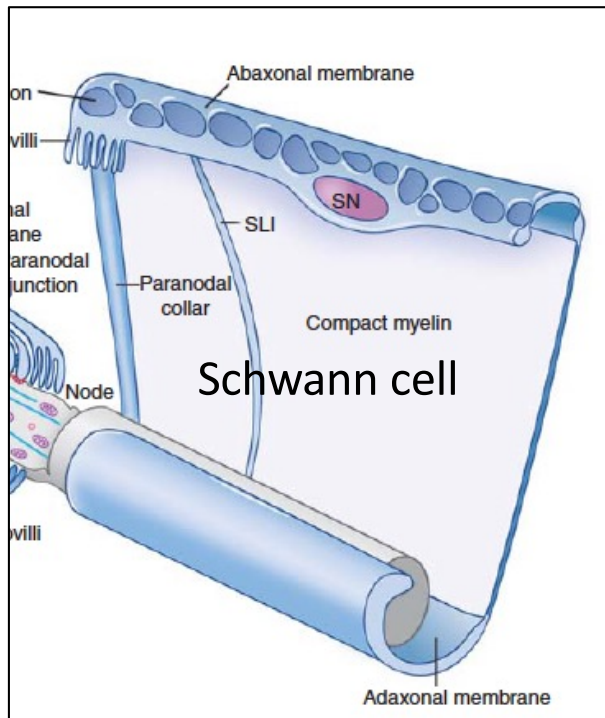


Le **cellule di Schwann** formano la mielina dei neuroni periferici

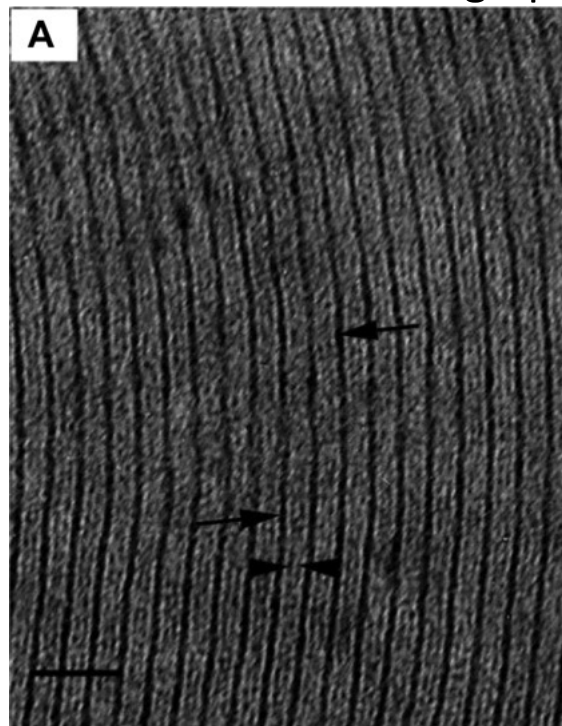
Schwann cells

Schwann cells play:

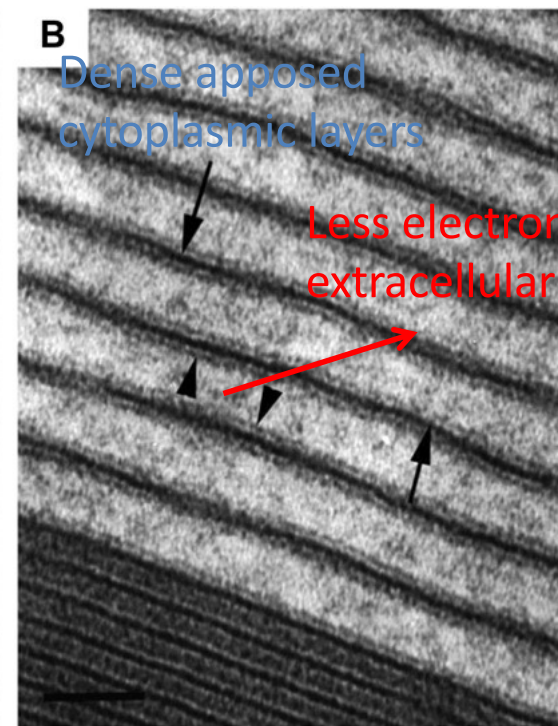
- a **supportive role** to fibers
- are involved in the **repair process**
- generate a **myelin sheath** by surrounding axons with multiple layers of compacted plasma membrane (lipid and protein lamellae)



Electron micrograph- Transverse section

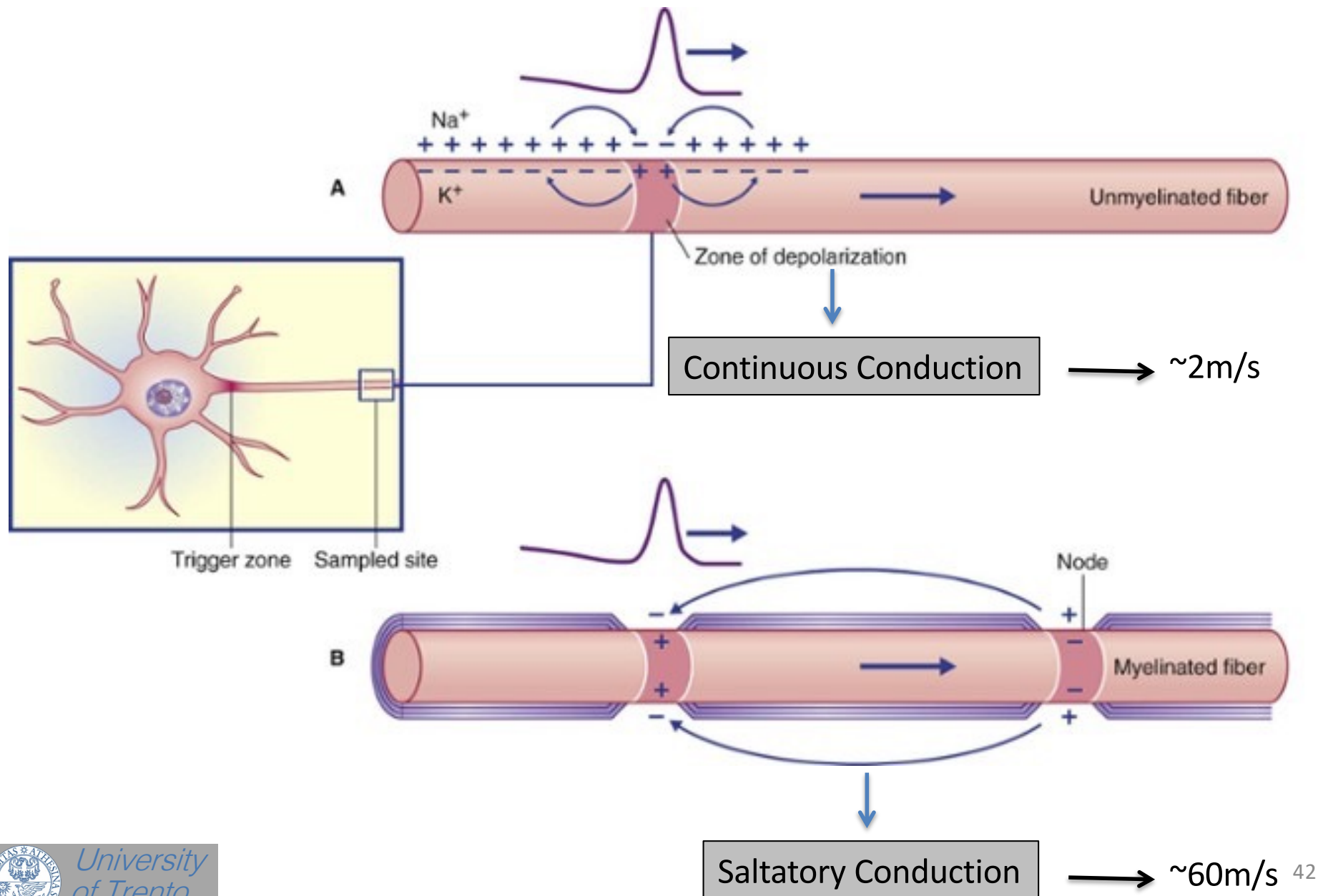


Normal myelin

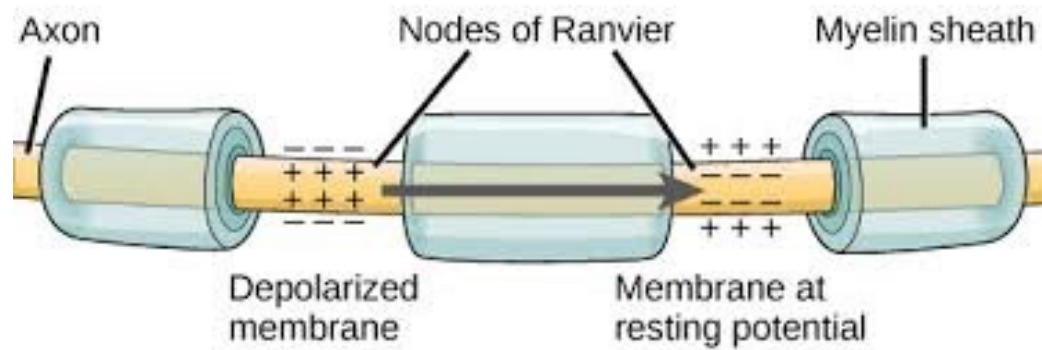


Widely spaced myelin in patients with IgM paraproteinemic neuropathy

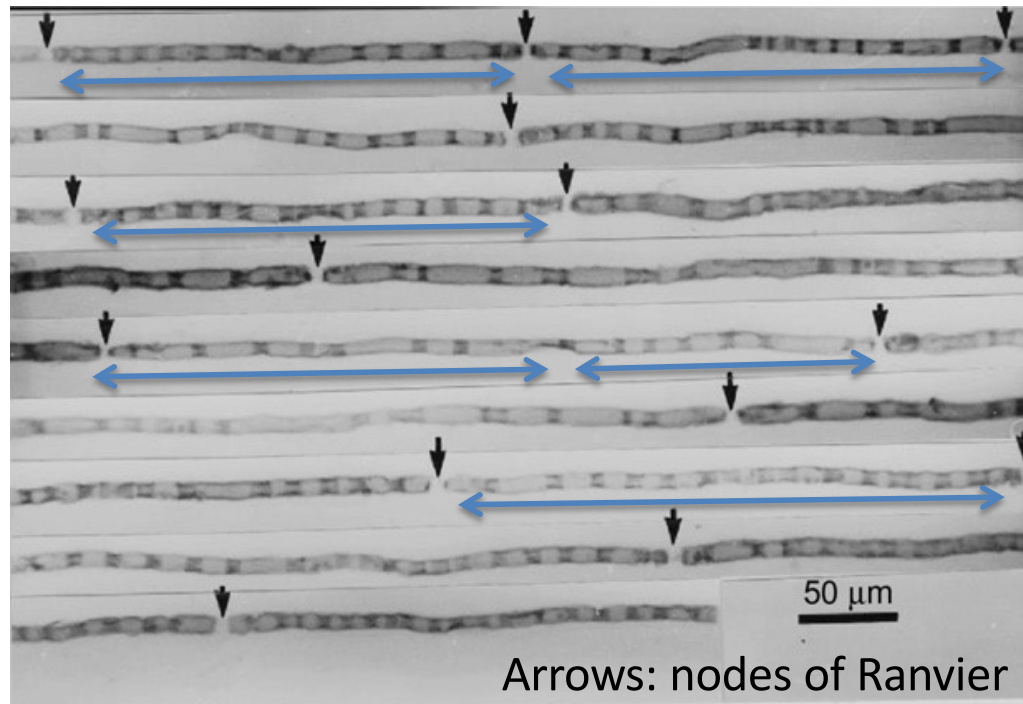
Impulse Propagation



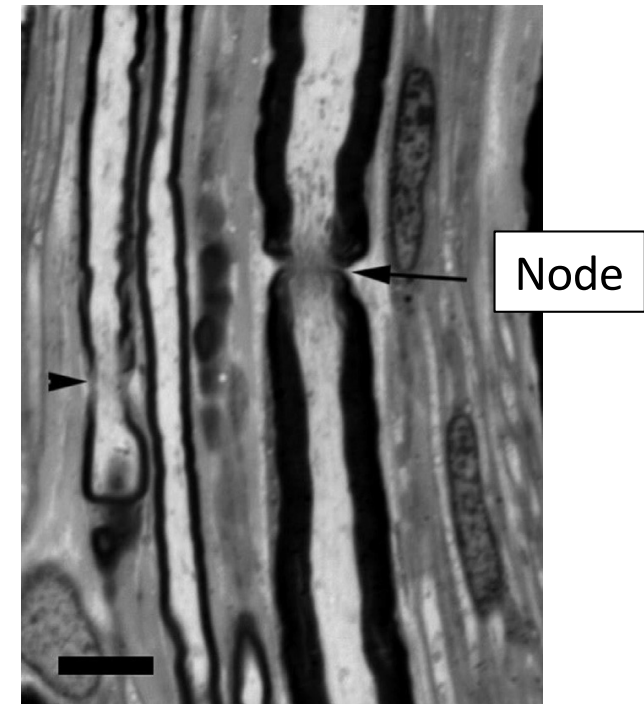
Multiple Schwann cells lie along myelinated fibers



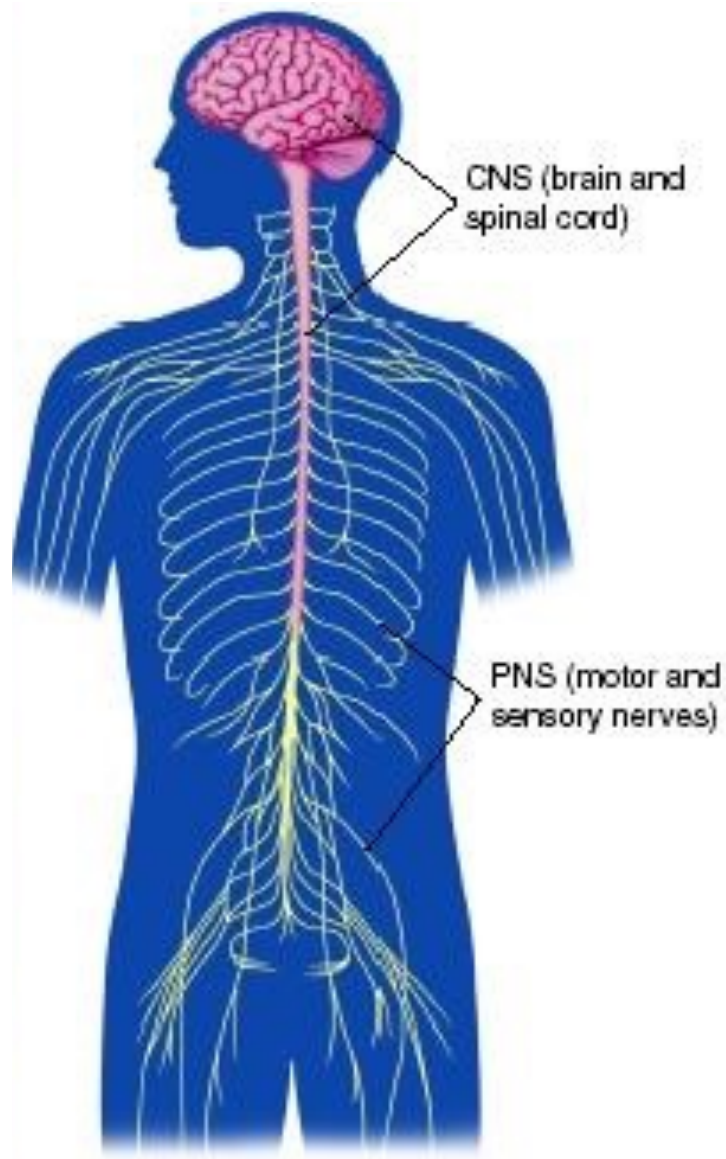
Sodium channels are concentrated at nodes of Ranvier (depolarization)



Internode: region between nodes



Components of the NS



CNS

Brain
Spinal Cord

PNS

Cranial nerves
Spinal Nerves
Autonomic Nerves and Ganglia
(including ENS)

NS

CNS

Brain protected by the **skull**
Spinal Cord protected by **vertebrae**

Brain has a **Blood-Brain Barrier (BBB)**
Spinal cord has **Blood-Spinal Cord Barrier (BSCB)**

Little regeneration

Specific immunitary defense
(i.e. microglia..)

PNS

PNS is not protected by
bones

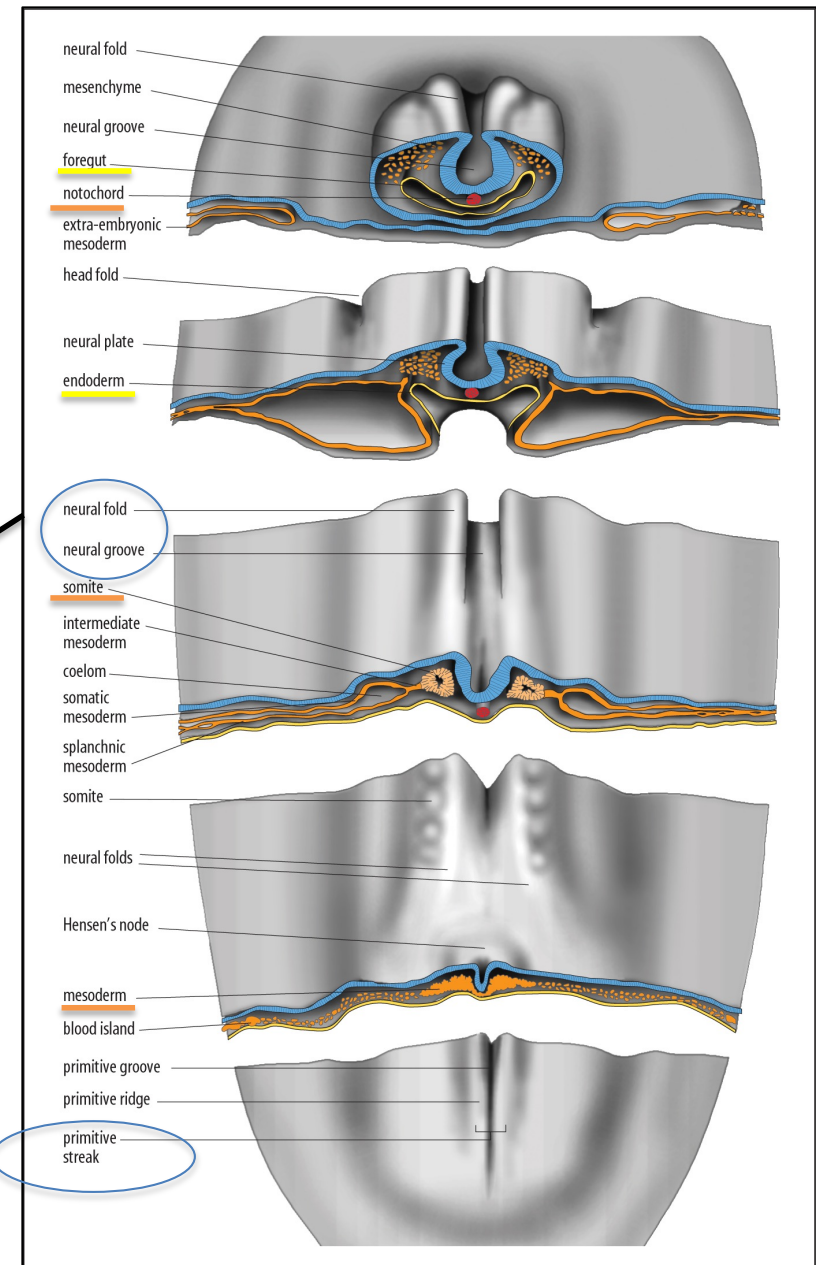
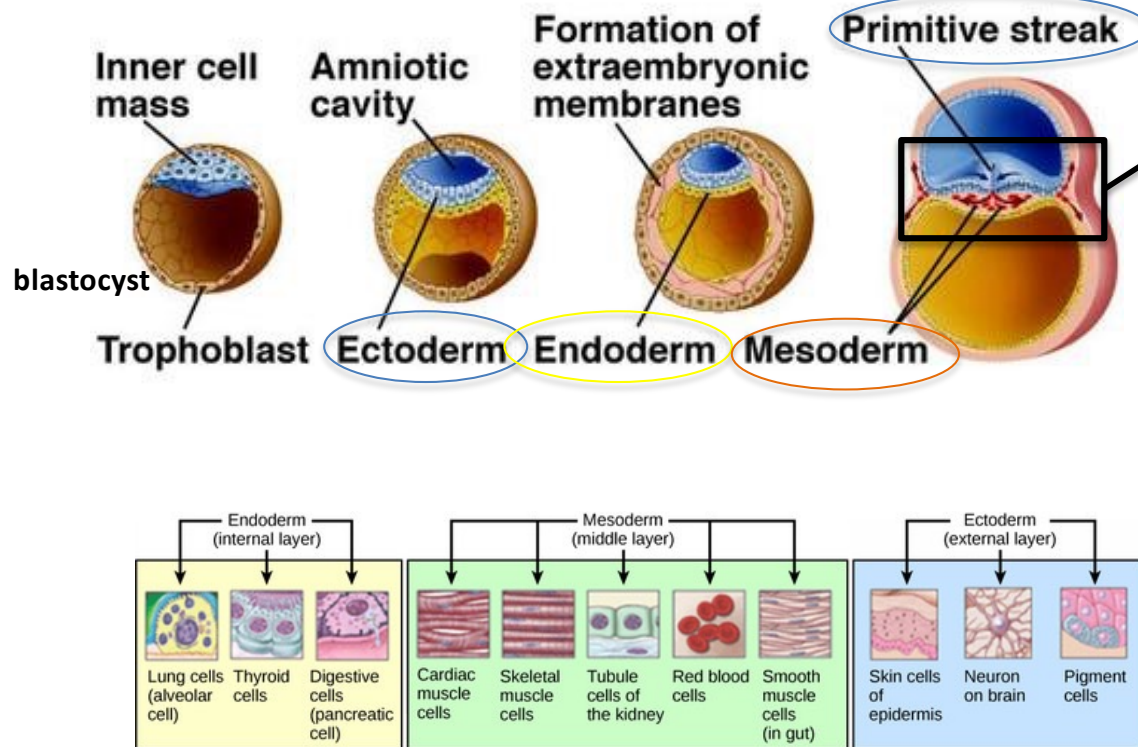
PNS has a less selective
Blood-Nerve Barrier

Higher regenerative
potential



Top view of an Embryo

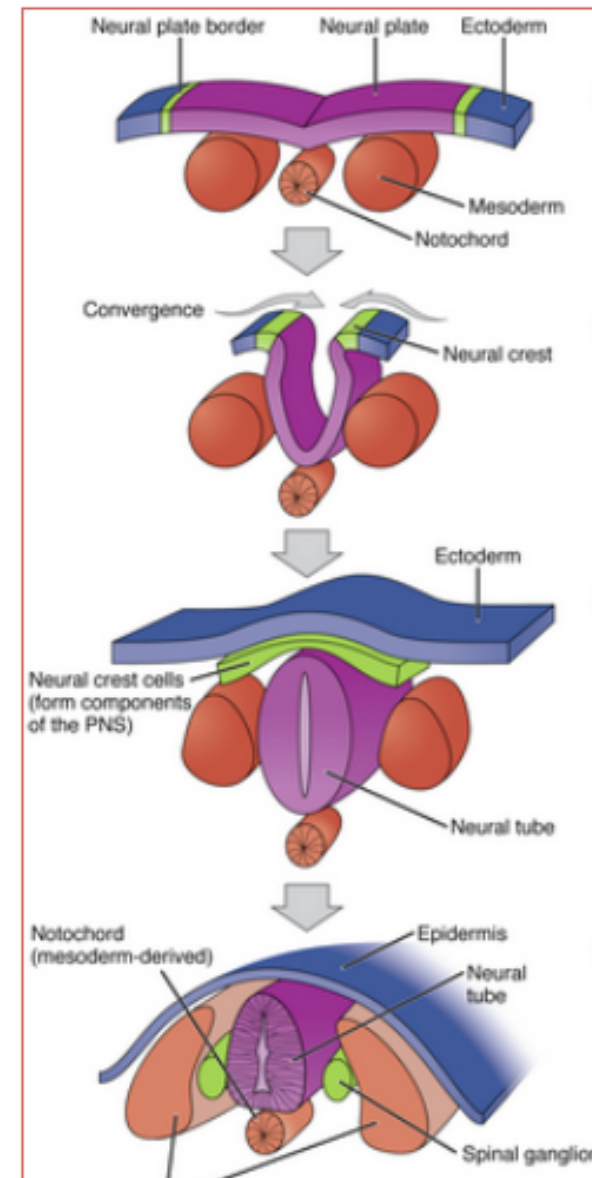
Gastrulation—Mammal



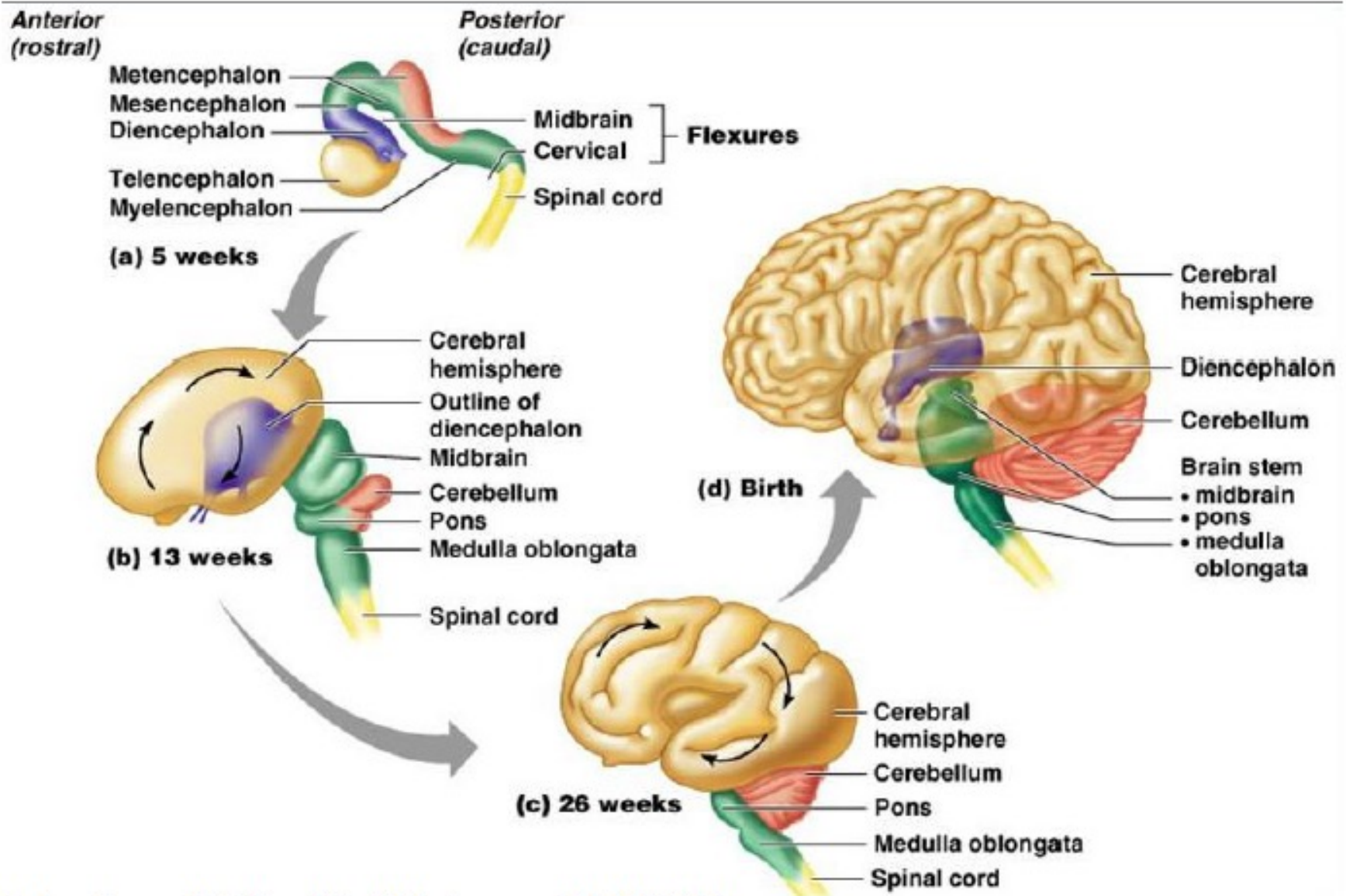
Lo sviluppo del sistema nervoso centrale

■ *Derivatives of Ectoderm: Nervous System*

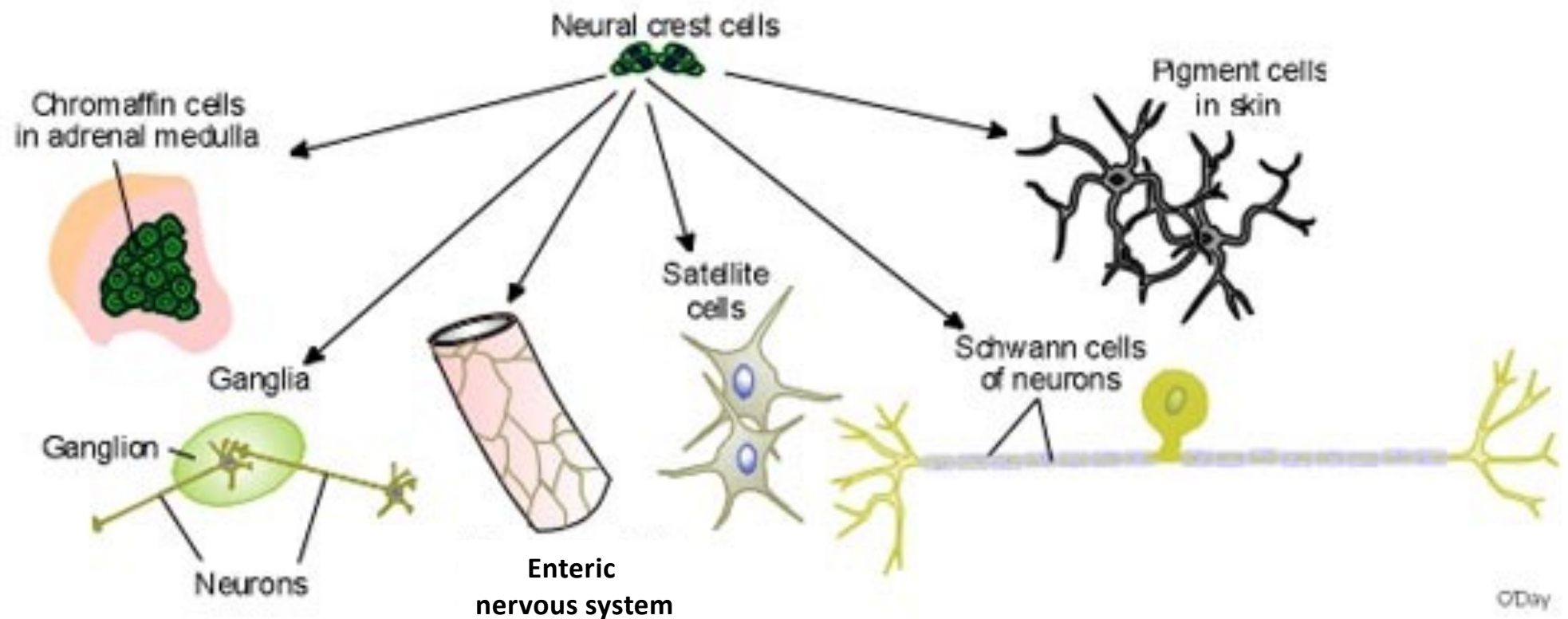
- The brain, spinal cord, and outer epithelial structures develop from the ectoderm
- Ectoderm thickens to form a *neural plate just above the notochord*
 - Edges of neural plate fold to create elongated *hollow neural tube*
 - Anterior end enlarges and forms the *brain and cranial nerves*
 - Posterior end forms the *spinal cord* and *spinal motor nerves*



Lo sviluppo del sistema nervoso



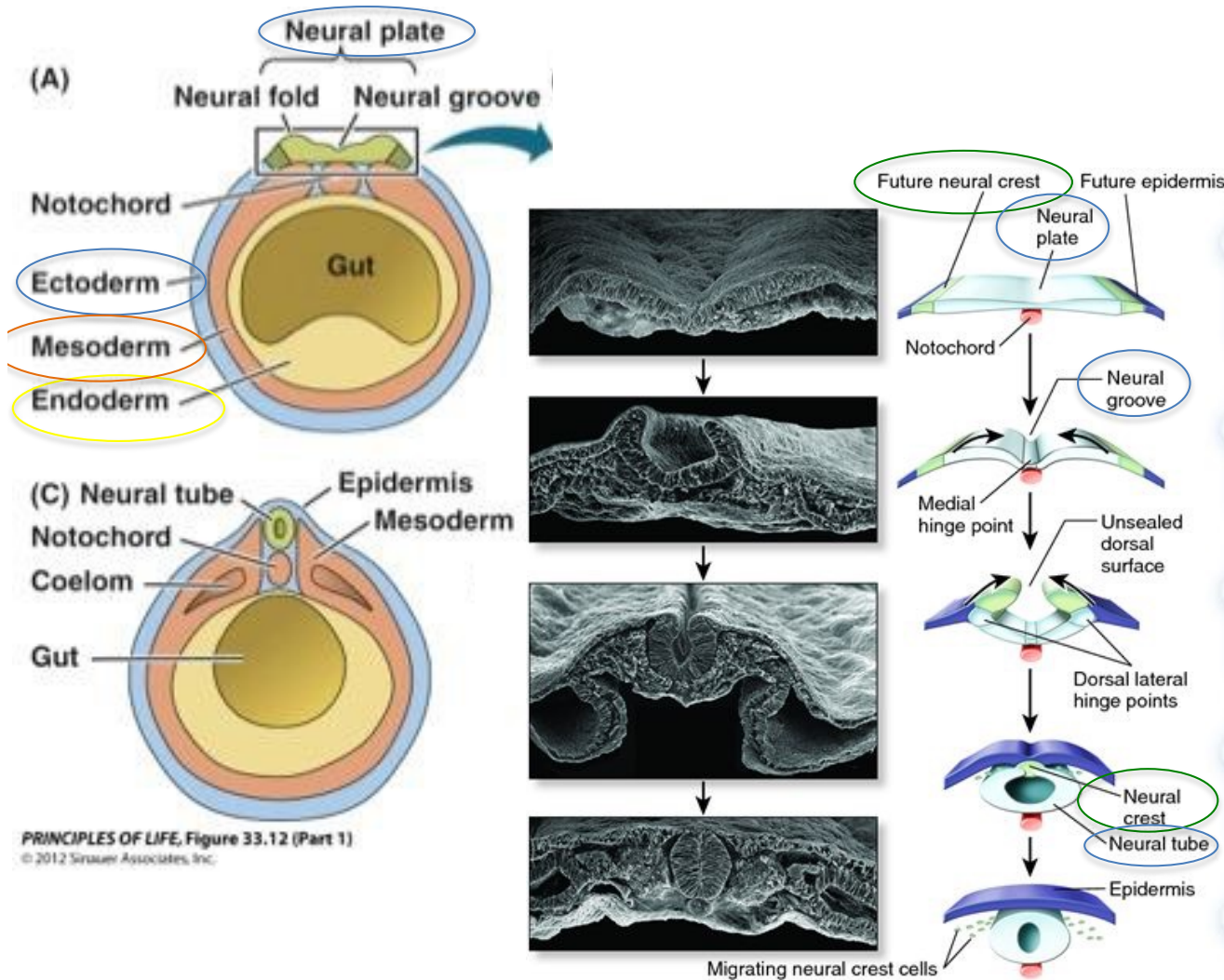
Most of the components of the PNS derive from Neural crest cells



Post-ganglionic autonomic neurons

Sensory neurons

Neural crest-development



1 Thickening and elongation:
Ectoderm over the notochord thickens to form the neural plate, which elongates. Adjacent regions will form a neural crest and epidermis.

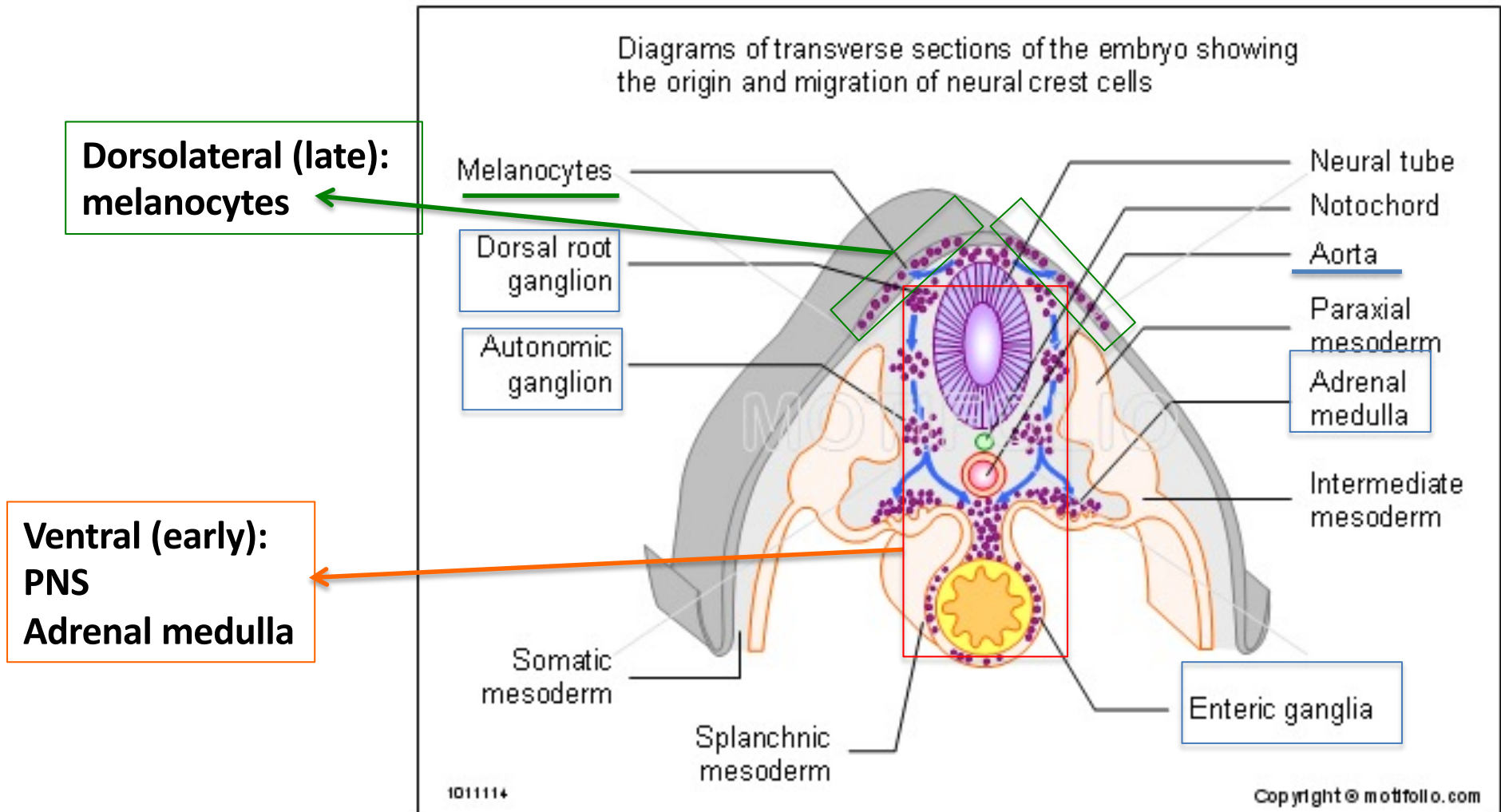
2 Folding:
Cells along the medial hinge point undergo apical constriction, causing the formation of a neural groove.

3 Convergence:
Two dorsal lateral hinge points undergo apical constriction to produce a tubelike structure.

4 Fusion:
The dorsal-most cells on each side of the neural tube begin to fuse. Epidermis also fuses dorsal to the neural tube.

5 Neural crest cells migrate away as fusion is completed.

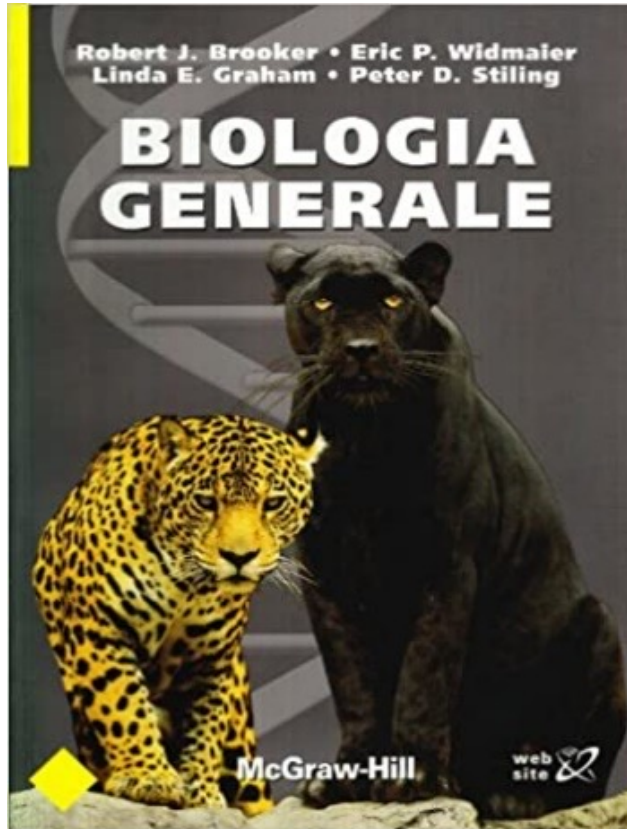
Ventral versus dorsolateral migration



SUMMARY

- 1. Le cellule del sistema nervoso centrale: i diversi tipi di neuroni, caratteristiche morfologiche e funzionali, le cellule gliali (astrociti e oligodendrociti)**
- 2. Il potenziale di membrana a riposo: come viene stabilito, la funzione dei canali ionici e le pompe ATP-asiche**
- 3. Cambiamenti nel potenziale di membrana: depolarizzazione, iperpolarizzazione e ripolarizzazione.**
- 4. Accenni allo sviluppo del sistema nervoso: dalla neurulazione alla formazione del cervello adulto, cellule della cresta neurale e sistema nervoso periferico**

Bibliografia



Capitolo 24



Capitolo 22